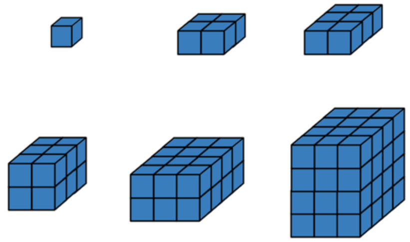
Unit 8 Overview: Surface Area and Volume

Unit Narrative

In this unit, students build on what they learned about area in Unit 7 and prior knowledge of volume of rectangular prisms from Grade 5 as they move into three-dimensional figures. This unit focuses on surface area and volume of rectangular prisms and pyramids.

Section 1 of this unit explores volume of rectangular prisms. In 5th grade, students found volume by packing three-dimensional figures with unit cubes. In this section, students will expand their understanding of volume by finding volume using several different methods, including finding volume by packing cubes with fractional edges and using the formula $Volume = length \times width \times height$ and $Volume = (area\ of\ base) \times height$. Students also apply these formulas to find a missing dimension given the volume.



In Section 2, students use prior knowledge on area to find the surface area of prisms and pyramids as well as to compare and connect the concepts of volume and surface area. This section connects the concepts of nets with that of composite figures and provides opportunities for students to draw and label nets of rectangular prisms and pyramids. Students will also find the surface area of rectangular prisms and pyramids when given a net or the figure.

Unit At-A-Glance

Section 1: Volume	Benchmarks Addressed	Lesson 1	Volume of Right Rectangular Prisms
	MA.6.GR.2.3	Lesson 2	Volume of Rectangular Prisms
		Lesson 3	Finding the Missing Dimension of a Rectangular Prism
Section 2: Surface Area	Benchmarks Addressed	Lesson 4	Nets of Rectangular Prisms and Pyramids
	MA.6.GR.2.3 **MA.6.GR.2.4**	Lesson 5	Surface Area of Rectangular Prisms and Pyramids
		Lesson 6	Finding Surface Area and Volume

Differentiation

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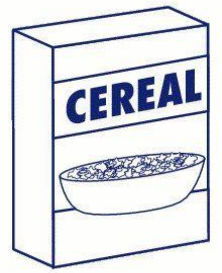
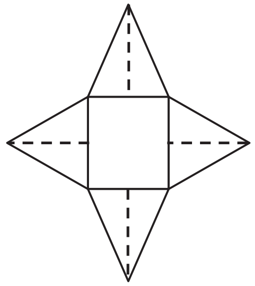
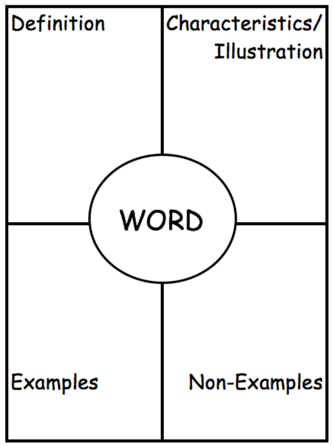
Engagement The "WHY" of learning	• Invite students to bring in real-life examples of three-dimensional figures, such as shoe boxes, empty milk cartons, or cereal boxes, to be used for nets and volume throughout the unit. • Scaffold the problem-solving process and encourage independence and confidence by having students describe or depict their methods for solving volume and surface area problems. 
Representation The "WHAT" of learning	• Have students use different colors to identify various characteristics of the shapes, such as using one color to shade in the bases and using another color to shade in the lateral faces or different colors for length, width, and height. • Utilize various representations of three-dimensional figures by having students create physical models to represent three-dimensional figures. To represent volume, use isometric dot paper to draw three-dimensional figures on paper or provide unit cubes to aid reasoning about fractional side lengths and volume. Creating nets from paper, cutting out the two-dimensional shapes as faces, and taping or gluing the nets together can help students connect their understanding of area to surface area of three-dimensional figures. 
Action & Expression The "HOW" of learning	• Consider including movement in the Warm-Up sections in this unit, like Four Corners routines for Would You Rather and Which One Doesn't Belong? Warm-Up activities in lessons 5 and 6. Using the corners of the classroom, post pictures of the pattern/pictures, one per corner, and elicit student responses by having them move to the corner they choose as their response. Students can clarify their thinking with peers who similarly chose the same corner or try to justify their reasoning to others in different corners.

ELL Information

In this unit, students will be using key terms related to geometry, including recognizing the meaning of words like “volume” and “surface area” in real-world scenarios. To solidify understanding of these words in context, have students create Frayer Models for the words in this unit and have them available as a reference throughout the unit.

- A Frayer Model (similar to the four-square graphic organizer) is a helpful graphic organizer because it allows students to see terms in multiple ways. For this section, students will be using many words that connect and build upon geometric understanding of two-dimensional and three-dimensional figures.
- Provide the definition but have students work in pairs to illustrate the word (including the shape/figure and the net if applicable), list examples, and list nonexamples.
- For examples and nonexamples, encourage students to consider this as a work in progress and to add throughout the section as they are presented.
- Consider creating a Frayer Model for each term from the Glossary by Section lists (below).



All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”

- Cube
- Prism (right)
- Rectangular prism
- Net
- Pyramid (regular)
- Rectangular pyramid

Additional Differentiated Support

Scaffolding Resources

Graphic organizers can be used in Section 2 to scaffold students as they analyze nets and prisms to determine missing dimensions, find lateral surface area, calculate surface area, and find volume. The example shown from Lesson 5 helps students organize needed information and utilize formulas for area and surface area.

To support these skills, a graphic organizer can help students visually and clearly see what information is needed and the process for finding surface area or volume. This strategy may be beneficial for all learners, as it also helps develop students' abilities to recognize what information is necessary and what information is not needed when solving problems, by organizing the information in a logical process that can be followed as a problem-solving approach.

Side	Formula	Area
Rectangular base	$A = l \cdot w$	$A =$ _____ $\cdot$ _____ $A =$ _______
Triangular Face 1	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot$ _____ $\cdot$ _____ $A =$ _______
Triangular Face 2	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot$ _____ $\cdot$ _____ $A =$ _______
Triangular Face 3	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot$ _____ $\cdot$ _____ $A =$ _______
Triangular Face 4	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot$ _____ $\cdot$ _____ $A =$ _______
		Surface Area:

On Ramp

On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills:

- Calculating Volume
- Classifying Quadrilaterals
- Creating Composite Shapes
- Finding Area of Figures Made from Unit Squares
- Finding Area of Rectilinear Shapes
- Using the Area Formula for Rectangles

Section Overview

Section 1: Volume (Lessons 1-3)

Benchmarks Addressed	Learning Targets	
MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula. • *Clarification 1: Problem types include finding the volume or a missing dimension of a rectangular prism.*	Lesson 1	• I can pack a rectangular prism using a cube with fractional edges.
	Lesson 2	• I can find the volume of a rectangular prism.
	Lesson 3	• I can find the value of a missing dimension of a rectangular prism.

Section 2: Surface Area (Lessons 4-6)

Benchmarks Addressed	Learning Targets	
MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula. • *Clarification 1: Problem types include finding the volume or a missing dimension of a rectangular prism.* **MA.6.GR.2.4** Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's nets. • *Clarification 1: Instruction focuses on representing a right rectangular prism and right rectangular pyramid with its net and on the connection between the surface area of a figure and its net.* • *Clarification 2: Within this benchmark, the expectation is to find the surface area when given a net or when given a three-dimensional figure.* • *Clarification 3: Problems involving right rectangular pyramids are limited to cases where the heights of triangles are given.* • *Clarification 4: Dimensions are limited to positive rational numbers.*	Lesson 4	• I can draw the net of a rectangular prism or rectangular pyramid. • I can relate a net to its three-dimensional shape. • I can relate a rectangular prism or rectangular pyramid to its net.
	Lesson 5	• I can find the surface area of a rectangular prism using a net or the figure. • I can find the surface area of a rectangular pyramid using a net or the figure.
	Lesson 6	• I can find the surface area of right rectangular prisms and right rectangular pyramids. • I can find the volume of right rectangular prisms.

Vertical Alignment

Grade 6 Accelerated Unit 8
Surface Area and Volume

Grade 7 Accelerated Unit 9
Surface Area and Volume

Prior Course(s)	Course Level	Future Course(s)
MA.5.GR.3.1 Explore volume as an attribute of three-dimensional figures by packing them with unit cubes without gaps. Find the volume of a right rectangular prism with whole-number side lengths by counting unit cubes. **MA.5.GR.3.2** Find the volume of a right rectangular prism with whole-number side lengths using a visual model and a formula.	**MA.6.GR.2.3** Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula. **MA.6.GR.2.4** Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's nets.	**MA.7.GR.2.2** Solve real-world problems involving surface area of right circular cylinders. **MA.7.GR.2.3** Solve mathematical and real-world problems involving volume of right circular cylinders.

8.1 Volume of Right Rectangular Prisms

Lesson Introduction

Benchmark	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula.	Learning Target	• I can pack a rectangular prism using a cube with fractional edges.

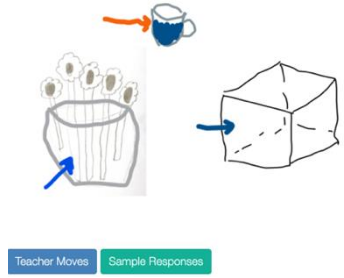
	Building On	Working Toward
Benchmark Coherence	MA.5.GR.3.3 Solve real-world problems involving the volume of right rectangular prisms, including problems with an unknown edge length, with whole-number edge lengths using a visual model or a formula. Write an equation with a variable for the unknown to represent the problem.	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

Essential Questions	• Why is volume measured in cubic units? • What is a unit cube? • How can you use a unit cube to find the volume of a rectangular prism? • What is the formula to find the volume of a rectangular prism?
Lesson Narrative	In this lesson, students investigate finding the volume of a right rectangular prism with fractional edge lengths by packing prisms with unit cubes of the appropriate unit fraction edge lengths. Students discover that finding the volume using this method produces the same result as finding the volume with the formula. The lesson begins with an exploration on packing rectangular prisms with cubes, but the bulk of the lesson focuses on guided instruction and practice in finding volume of right rectangular prisms and writing appropriate units.

Component Summary	8.1.1 Warm-Up: (~5 min.)	Representations and descriptions of volume *(SE p. 3)*	8.1.4 Guided Instruction (20-25 min.)	Students calculate the volume of rectangular prisms with unit cubes and compare results with the volume formula. *(SE p. 6)*
	8.1.2 Exploration (5 min.)	Explore volume and dimensions of the unit cube *(SE p. 4)*	8.1.5 Your Turn! (5-10 min.)	Independent practice finding the volume of a rectangular prism with fractional side lengths *(SE p. 10)*
	8.1.3 Exploration (10-15 min.)	Explore volume of cubes with fractional side lengths by comparing to the unit cube *(SE p. 5)*	8.1.6 Wrap-Up (~5 min.)	Students match prisms with correct dimensions and volume *(SE p. 10)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Cube <u>Additional Terms:</u> • Volume • Length, Width, Height • Unit cube	• (Optional) Unit cube manipulatives • (Optional) Paper • (Optional) Rulers • (Optional) Scissors	• There is a Desmos activity builder to accompany this lesson. If using the Desmos version of the activity, review slides ahead of time.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students with gaps in spatial awareness may struggle with composing and decomposing rectangular prisms into fractional cubes.
Students may struggle to differentiate between two-dimensional faces and three-dimensional cubes.</td><td>Facilitate class discussion on 8.1.3 Exploration question 3, which could prove helpful for students as they visualize decomposing a larger rectangular prism into smaller cubes.
Consider having physical unit cubes available for students to manipulate into larger prisms.
The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions. Once students understand conceptually how to break a cube into fractional edges, then the mathematics makes sense. Utilizing the Desmos activity will help students visualize the concepts. Students may go through the Desmos activity faster than the workbook lesson. Determine if students should record their answers from the Desmos activity into the workbook.
If time is an issue, then assign 8.1.5 Your Turn! for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
8.1.3 Exploration intentionally sets the stage for students to think about the relationship between unit cubes and the volume of a larger prism. This low-stakes task makes the exploration accessible to all students and provides opportunities for students to share what they notice and wonder about volume and unit cubes.</td><td>MA.K12.MTR.2.1: Multiple Representations
In this lesson, students are actively engaged building multiple representations to demonstrate understanding in more than one way, including through modeling and manipulatives. Utilize 8.1.3 Exploration and 8.1.4 Guided Instruction questions to position students to recognize volume with a visual representation and the formula by connecting the process of packing prisms with cubes and the volume formula.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson focuses heavily on spatial reasoning and promotes visual representations of volume of rectangular prisms using unit cubes. Providing physical manipulatives or tools to draw visual models in several lesson activities (8.1.3 Exploration, 8.1.4 Guided Instruction, and 8.1.5 Your Turn!) can allow additional entry points for kinesthetic and visual learners.

Additionally, students can engage with each other synchronously or asynchronously using the accompanying Desmos Activity Builder for this lesson. Consider leveraging the digital version of this lesson before, during, or after class for additional entry points to the learning.

</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

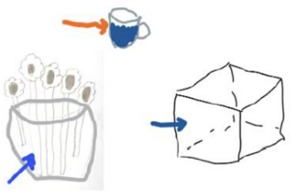
8.1.1 Warm-Up: Guess the Word

Component Overview: *Students observe three visual representations of volume and choose statements that best describe volume. Consider using Notice and Wonder routines as a group and write students' responses in a shared classroom space. Look for general themes from their responses, like volume, area, 3D object, and/or content in an object. Prompt responses within these themes and ask students to briefly elaborate.*

8.1.1 Warm-Up: Guess the Word

Micah and his family were playing a game in which each person chooses a card with a word written on it. Once a person chooses, they have one minute to draw a picture of the word. Then, his team guesses the word.

Micah chose the word "volume". His drawing is shown.



1. Choose the statement(s) that best describe volume. Select all that apply.

   - ☐ the sum of the length of all the edges of a three-dimensional object
   - ☐ the distance around a two-dimensional shape
   - ☐ the amount of three-dimensional space enclosed within a container
   - ☐ the total area of the surface of a three-dimensional object
   - ☐ the formula length times width times height

As students look at the math representations that Micah drew to represent volume (which show the space inside a cup, flowerpot, and cube), encourage mathematical reasoning about volume by asking questions like:

- *What do you notice about the drawings? What do you wonder?*
- *What do you think each drawing represents? What is the same or different about these objects in the real-world?*
- *Which drawings represent volume? How can you tell?*

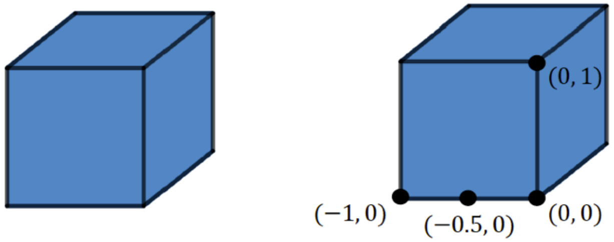
Early finishers can create their own drawing to represent volume, then compare their drawing to Micah's or another student's work. This activity can also be extended by having students identify what mathematical concept is described by the three incorrect answers in question 1.

8.1.2 Exploration: The Unit Cube

Component Overview: *Students make observations about the unit cube's dimensions and explore its volume. Consider having partners complete questions 1 and 2 for about five minutes while writing students' observations and questions on a shared classroom space, then regroup and revisit questions from question 1 at the end of 8.1.2 Exploration.*

8.1.2 Exploration: The Unit Cube

1. Two images of the same cube are shown. The image on the right shows the coordinates of some of the edges of the cube.



Share three things you know, two things you are not sure of, and one thing you have a question about.

I Know…	I'm Not Sure Of…	A Question I Have…
Answers will vary. A cube is a 3D shape. A cube is made of squares. A cube has six sides.	*Answers will vary. How to find the length of a side. Which points to use to find the side lengths.*	*Answers will vary. How can I find out how much space is inside the cube?*

2. This cube is known as a unit cube because each dimension (length, width, and height) is 1 unit.

What is the volume of the cube? Include the units for the cube's volume.

1 unit^3

A ***cube*** is rectangular prism with six congruent square faces.

For question 1, consider highlighting the most common responses for each category, in particular the questions students have. Consider posing some questions or aspects students are unsure about to the whole group to gather students' initial thoughts, then prompt students to attempt answering the questions at the end of the 8.1.2 Exploration.

Emphasize what is considered a unit cube and why it is a unit cube. This helps students then relate to fractional cubes. If manipulatives are available, then consider showing one cube as the unit cube.

Scaffold thinking for question 1 by prompting students to consider:

- *How can the coordinates help us to better understand the dimensions of the cube?*
- *What is the distance between coordinates? What does that tell you about the distance of the cube's edges?*
- *Could a different set of coordinates produce a unit cube? Why or why not?*

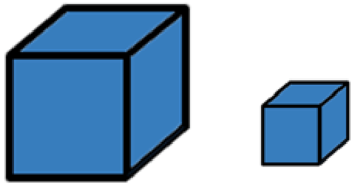
8.1.3 Exploration: Drawing Fractional Cubes

Component Overview: *Students explore how to decompose the unit cube into cubes with fractional side lengths of $\frac{1}{2}$ and discover different ways to visualize the total number of these cubes contained within a unit cube. Consider using Notice and Wonder routines as a class with questions 1-3 to highlight important concepts related to volume, then have partners continue working together to complete question 4.*

8.1.3 Exploration: Drawing Fractional Cubes

1. Two cubes are shown.

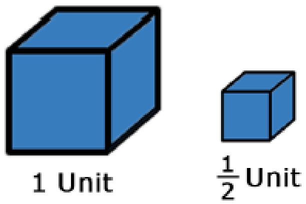
The cube on the left is a unit cube. The cube on the right has different dimensions.



Share an observation and something you wonder about this new cube.

Observation	Wondering
Answers will vary. The cube on the right is smaller than the cube on the left. Both cubes have the same amount of sides.	*Answers will vary. How many small cubes can fit in the big cube?*

A unit cube and a half-unit cube are shown.



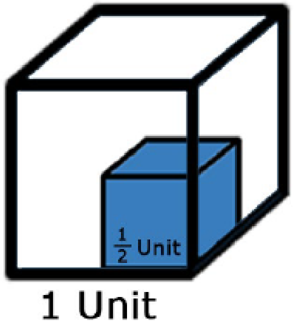
Make a guess of how many cubes with a length of $\frac{1}{2}$ unit, width of $\frac{1}{2}$ unit, and height of $\frac{1}{2}$ unit will fit in the 1-unit cube.
Answers will vary. 2, 4, 8

Notice and Wonder
Use these first two questions as opportunities to have students discuss the relationship between side lengths and volume. Some students may recall the volume formula (length×width×height).

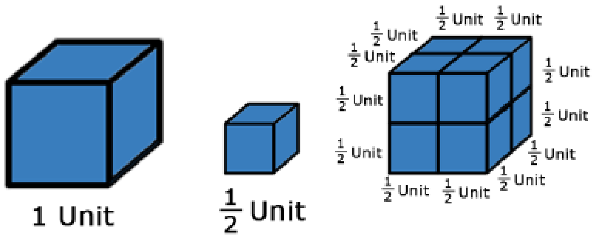
- *How many half unit cubes did you guess would fit in a unit cube? How did you make your guess?*
- *How much larger do you think the volume of the unit cube is compared to the half unit cube?*

2. The figure shows a half-unit cube inside a unit cube.

Explain how the figure helps you to visualize how to pack cubes that have a fractional length.
Answers will vary.
It looks like I make two layers of 4 for a total of 8 half-unit cubes.



3. Three figures are shown.



a. Explain how the figures help you visualize how to pack cubes that have a fractional length.
Answers will vary.
I was able to see how the cubes were half of the side lengths. It reminded me of a 2x2 Rubik's Cube.

b. To find the number of cubes whose dimensions are $\frac{1}{2}$ unit that fit into a cube whose dimensions are 1 unit, Derek wrote 1 as $\frac{2}{2}$ so he could see that $\frac{1}{2}+\frac{1}{2}$ would add to length of $\frac{2}{2}$ or 1. Complete his explanation.
Since $\frac{1}{2}+\frac{1}{2}=\frac{2}{2}$, or 1, I need __2__ cubes for the length, __2__ cubes for the width, and __2__ cubes for the height.
Since __2__ $\times$ __2__ $\times$ __2__ $= 8$ I know that I need 8 cubes with dimensions $\frac{1}{2}$ unit to fit into one 1-unit cube.

How do you know there are four cubes in each level?

Consider allowing students who struggle with spatial reasoning to use cube manipulatives to physically pack a cube or build a side-by-side view of a unit cube and a fraction cube. If physical cubes are unavailable, students can also use the Desmos activity to virtually pack the cube.

Students who finish early could extend their thinking by answering the following prompts:

- *Would any other cube size fit inside the unit cube?*
- *Choose a different dimension cube (e.g., $\frac{1}{3}$ unit) and find how many of those smaller cubes can be packed into the unit cube.*
- *Can you come up with a pattern or rule to find the total number of cubes in a unit cube? What is the pattern or rule? How do you know it always works?*

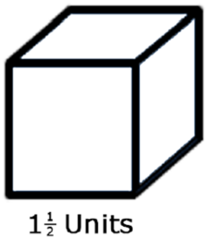
8.1.4 Guided Instruction: Volume with Fractional Cubes

Component Overview: *This section guides students through the process of decomposing prisms into cubes to calculate and recognize a representation of volume. Students confirm this method by applying the volume formula for a rectangular prism ($V = lwh$). Consider guiding students through each question in this section as a class, pausing to discuss questions after each question.*

8.1.4 Guided Instruction: Volume with Fractional Cubes

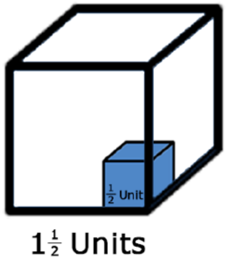
The cube shown is a scale model of a moving box.

1. The edges of the box are $1\frac{1}{2}$ feet long. How many cubes, whose dimensions are $\frac{1}{2}$ feet, can fit inside each dimension of the moving box?
   3



For this activity, consider demonstrating directly how to sketch the cubes inside the larger prism.
Have students work in partners to check one another's sketches. It is important that students do not overcomplicate the sketching process. Instead, sketching helps them to quickly see how many cubes are in a layer. Some students may find a rule is helpful while sketching.

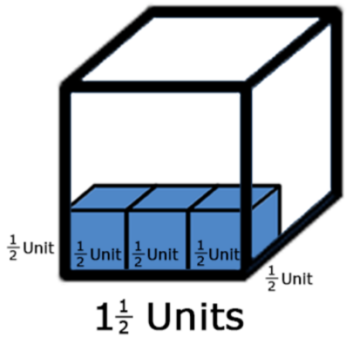
2. Sketch the cubes that can fit inside each dimension of the moving box.
   Check student drawings for understanding. Each side of the cube should be divided into 9 squares.



Prompt students to consider how the cubes relate to volume by asking:

- *How many squares can you see on each side of the cube?*
- *Is this the same as the total number of cubes packed inside? Why or why not?*

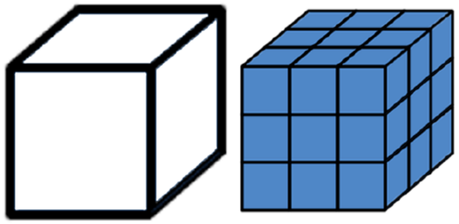
3. Since 3 cubes, whose dimensions are $\frac{1}{2}$ unit, will fit inside each dimension, how many of these cubes would be needed to pack the moving box?
   27



Consider asking students to discuss multiple pathways to discover that 27 cubes are needed to pack the box. Allow extra time for this discussion because it is important for students to understand how to arrive at a total number of cubes.

8.1.4 Guided Instruction: Volume with Fractional Cubes (continued)

4. This cube is a scale model of a moving box. The edges of the box are $1\frac{1}{2}$ feet long.



Since there are 27 cubes whose dimensions are $\frac{1}{2}$ feet that will fill the moving box, the volume of the moving box can be found by multiplying the volume of the fractional cube by 27. What is the volume of a cube whose dimensions are $\frac{1}{2}$ feet?

$\frac{1}{8} ft^3$

To scaffold thinking for question 4, encourage students to revisit 8.1.3 Exploration, prompting them to consider:

- *What would the volume be for a cube whose dimensions are 1 foot? Why do we call it a unit cube?*
- *How many half cubes were able to fit inside one unit cube?*
- *What is the volume of a single half cube? How do you know?*

Ensure that students understand that any of the three dimensions could be labeled as length. The "why" is easily visualized by rotating a cube or rectangular prism so that the side that is on the bottom varies. Practice using various labels to help students relate to this flexibility.

8.1.4 Guided Instruction: Volume with Fractional Cubes (continued)

5. Since there are 27 cubes whose volume is $\frac{1}{8}$ cubic feet that will fill in the moving box, the volume of the moving box can be found by multiplying 27 and $\frac{1}{8}$ cubic feet. What is the volume of a cube of the moving box?
   $3\frac{3}{8} ft^3$

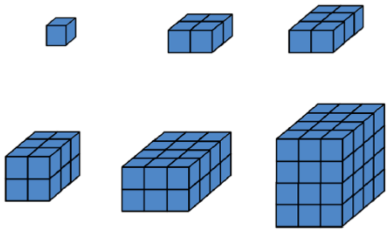
6. Find the volume of the moving box using the formula for volume of a rectangular prism or cube.

$$V = lwh$$

   Each dimension of the box is $1\frac{1}{2}$ feet.
   $V = lwh = \frac{3}{2} \times \frac{3}{2} \times \frac{3}{2} = \frac{27}{8} = 3\frac{3}{8} ft^3$

7. Does the volume you found using the formula match the volume you found by counting the number of fractional cubes?
   Yes

8. The images show different rectangular prisms composed of various numbers of cubes.



   a. What do you notice?
      Answers will vary. If I multiply the amount of cubes that make up each side length, it equals how many cubes are shown total.
   b. Explain how the dimensions of the prism can be determined using the cubes shown if each small blue cube is 1 unit cube.
      You can count how many cubes make up each side to use for length, width, and height to find the volume.

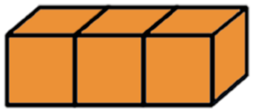
Strengthen student understanding of how to use the formula to find volume in questions 6 and 7 by asking:

- *What do $l, w,$ and h represent in the formula?*
- *What is the length of the cube? Width? Height?*
- *How do we substitute values into this formula?*

In question 8, position students to note the length, width, and height of the larger prism (in addition to the total number of cubes). This sets students up to be able to use these dimensions in calculating volume. Have students label the dimensions on the prisms.

8.1.4 Guided Instruction: Volume with Fractional Cubes (continued)

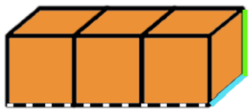
9. The diagram shows 3 cubes whose dimensions are $\frac{1}{4}$ inches.



Select the statements that correctly describe the volume of the diagram as shown.

- ☐ Each cube has a volume of $\frac{1}{4}$ cubic inches.
- ☐ Each cube has a volume of $\frac{1}{16}$ cubic inches.
- ☐ Each cube has a volume of $\frac{1}{64}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{4}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{16}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{64}$ cubic inches.

10. The same diagram of 3 cubes whose dimensions are $\frac{1}{4}$ inches is shown.



a. The dotted line represents the total length of the 3 cubes. How long, in inches, is the dotted line?

- Ⓐ $\frac{4}{4}$ inches
- Ⓑ $\frac{1}{4}$ inches
- Ⓒ $\frac{3}{4}$ inches
- Ⓓ $\frac{2}{4}$ inches

Consider allowing students to work in partner pairs for questions 9-11. Circulate the room and monitor for correct application of the volume formula. Student approaches might illuminate potential gaps (e.g., multiplying fractional side lengths to find volume).

As a visual entry point to question 9, consider prompting students to draw arrows to the diagram or to arithmetic work as evidence on why they know the statements are correct.

For question 10, deepen conceptual understanding by eliciting responses to the following:

- *Explain why the volume of each cube is $\frac{1}{64} in^3$.*
- *How can you find the volume of the entire prism once you know the volume of one cube?*

8.1.4 Guided Instruction: Volume with Fractional Cubes (continued)

b. The blue line represents the width of the 3 cubes.

What is the width, in inches?

Ⓐ $\frac{1}{4}$ inches
Ⓑ $\frac{2}{4}$ inches
Ⓒ $\frac{3}{4}$ inches
Ⓓ $\frac{4}{4}$ inches

c. The green line represents the height of the cubes.

What is the height, in inches?

Ⓐ $\frac{1}{4}$ inches
Ⓑ $\frac{2}{4}$ inches
Ⓒ $\frac{3}{4}$ inches
Ⓓ $\frac{4}{4}$ inches

d. Use the dimensions you found to determine the volume of all three cubes with the formula, $V = lwh$.

$V = 3 \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{3}{64} in^3$

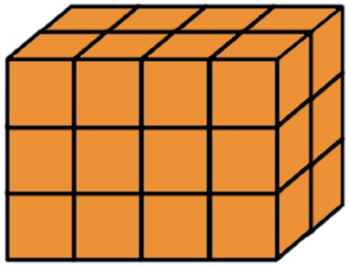
Finding volume using fractional unit cubes can be done by first finding the volume of one of the fractional unit cubes, and then multiplying that by the total number of fractional unit cubes.

For questions 10 and 11 and into the practice in the next section (8.1.5 Your Turn!), consider providing a template or graphic for students who may struggle with organization. The organizer can have prompts to substitute into the volume formula, reminders on how to multiply fractional values, and tips on converting from improper to mixed-number form.

8.1.4 Guided Instruction: Volume with Fractional Cubes (continued)

11. A rectangular prism is shown.

   a. How many unit cubes whose dimensions are $\frac{1}{4}$ inch fill up the rectangular prism?
   24



   b. Each unit cube whose dimensions are $\frac{1}{4}$ inch has a volume of $\frac{1}{64}$ cubic inches.

   Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.
   $\frac{3}{8} in^3$

   c. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$. Show your work and your answer.
   $V = (3)\frac{1}{4}(4)\frac{1}{4}(2)\frac{1}{4}$
   $V = \frac{3}{8} in^3$

For question 11a, if a student multiplied 24 by $\frac{1}{4}$ to find the volume of the prism, would the student arrive at the correct volume? Why or why not?

Encourage synthesis of the learning targets in this section by positioning students to recall the two strategies to finding volume. Consider using Turn and Talk routines to have students summarize the two strategies to finding the volume of rectangular prisms with fractional side edges.

- *Strategy 1: Choose a unit cube that fits into each dimension. Multiply the total number of cubes that pack the prism by the volume of the unit cube.*
- *Strategy 2: Apply the volume formula, $V = lwh$.*

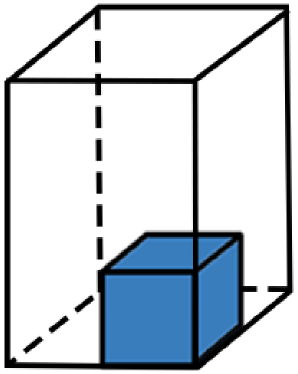
Additionally, consider having students share which strategy they prefer and why. Use this reflection to highlight strategic approaches to finding volume, including when it may be better to use the formula versus when it may be easier to use unit cubes.

8.1.5 Your Turn!

Component Overview: *Students work independently to calculate the volume of a rectangular prism with fractional edges by packing the prism with cubes. While some of the class completes this work independently, consider pulling a small group of students who may need additional support in using the strategies to find volume from the previous section.*

8.1.5 Your Turn!

1. A rectangular prism with one cube inside is shown.



   a. How many cubes will it take to fill the rectangular prism?
   12

   b. The rectangular prism has a length of $\frac{16}{3}$ centimeters (cm), width of $\frac{16}{3}$ cm, and height of 8 cm as shown.
   Use the number of cubes in each dimension to determine the dimensions of the fractional unit cube.
   height = $8 \div 3$, *length and width* = $\frac{16}{3} \div 2$

   c. Complete the statement.

   The unit cube shown has dimensions of $\frac{\boxed{8}}{3}$ cm. It will take __12____ of these unit cubes to fill the rectangular prism.

   d. Explain how to find the volume of the rectangular prism using the information from Part C.
   To find the volume of the rectangular prism using the information from Part C, find the volume of the unit cube by multiplying the dimensions 3 *times. Then multiply the volume of the unit cube by the number of cubes it will take to fill the rectangular prism.*
   $\frac{8}{3} \times \frac{8}{3} \times \frac{8}{3} \times 12 = 227\frac{5}{9} cm^3$

Students may still struggle with differentiating between the volume of a single unit cube and the volume of the entire prism. Consider using this section as an opportunity to review strategies to find the total number of cubes and strategies to determine the dimensions of the cube with a small group of students who need additional support.

Students who finish early can confirm their answer in question 1d by applying the volume formula and comparing solutions.

8.1.6 Wrap-Up: Ticket Out the Door

***Component Overview**: Students complete a match activity that assess their ability to identify the dimensions of a rectangular prism with fractional side lengths and calculate volume. This Wrap-Up is composed of three multi-step problems that require proficiency in multiplying fractional values, determining dimensions of a rectangular prism, and calculating the volume of a rectangular prism.*

8.1.6 Wrap-Up: Ticket Out the Door

1. Match each image with its corresponding dimensions and volume. Use the table below to record your answers.

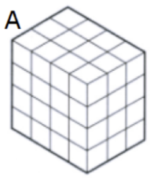
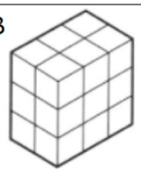
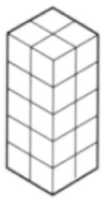
Image	Dimensions	Volume
A  The edges of each cube are $3\frac{1}{4}$ in. long.	D1 Length - 5 in. Width - 5 in. Height - 12.5 in.	V1 Volume $5\frac{1}{3}$ cubic in.
B  The edges of each cube are $\frac{2}{3}$ in. long.	D2 Length - 13 in. Width - 9.75 in. Height - 13 in.	V2 Volume 1,647.75 cubic in.
C  The edges of each cube are 2.5 in. long.	D3 Length - 2 in. Width - $\left(\frac{4}{3}\right)$in. Height - 2 in.	V3 Volume 312.5 cubic in.

Image	Dimensions	Volume
A	*D2*	*V2*
B	*D3*	*V1*
C	*D1*	*V3*

For students who typically require additional time to complete calculations, consider removing image C from the required task so students can focus on correctly calculating the volume for images A and B.

8.2 Volume of Rectangular Prisms

Lesson Introduction

Benchmark	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.	Learning Target	• I can find the volume of a rectangular prism.

Benchmark Coherence

Building On	Working Toward
MA.5.GR.3.3 Solve real-world problems involving the volume of right rectangular prisms, including problems with an unknown edge length, with whole-number edge lengths using a visual model or a formula. Write an equation with a variable for the unknown to represent the problem.	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

Essential Questions

- How can you find the volume of a rectangular prism?
- How is the method for calculating the area of a rectangle related to the method for calculating the volume of a rectangular prism?

Lesson Narrative

This lesson builds on students' experiences in packing rectangular prisms with cubes whose side lengths have rational values and introduces problem-solving with volume in real-world contexts. Students compare the two methods for calculating volume and discover a new volume formula ($V = Bh$), then explore real-world application problems with volume, including problems where smaller cube units are encased within a rectangular prism.

Component Summary

8.2.1 Warm-Up (~5 min.)	Investigate a box of cubes (*SE p. 11*)	8.2.2 Guided Instruction (10-15 min.)	Explore and review two methods to calculate volume of rectangular prisms (*SE p. 12*)
8.2.3 Guided Practice (10-15 min.)	Solve multi-step problems involving the volume of prisms (*SE p. 13*)	8.2.4 Your Turn! (5 min.)	Independent practice on solving real-world volume problems (*SE p. 14*)
8.2.5 Wrap-Up (~5 min.)	Assessment on calculating volume (*SE p. 15*)		

Resources

Glossary	Materials	Additional Preparation
<u>Key Terms:</u> • Volume <u>Additional Terms:</u> • Unit cube • Dimension, Base	• (optional) Chart paper • (optional) Markers • (optional) Unit cube or rectangular prism manipulatives • (optional) Four-Function calculator	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may struggle with visualizing the number of cubes packed within a prism and keeping track of the different values used throughout the problems.
• Students may struggle with operational fluency with fractions and mixed numbers.</td><td>• Virtual applets like Desmos can help students to visualize filling a rectangular prism with fraction unit cubes.
• Consider posting an anchor chart of fraction operations for students to reference throughout the lesson.
• Allowing students to use calculators during the lesson will help students focus on discovery and process of finding volume of a rectangular prism. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with positive decimals and positive fractions. For example, question 1 in 8.2.2 Guided Instruction could be a no calculator problem but question 2 in 8.2.2 Guided Instruction could be a calculator problem.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Number Talk
8.2.3 Guided Practice provides an opportunity to build computational fluency by encouraging students to compare different fractional unit lengths within a prism. Consider using Number Talk routines to facilitate number sense and reasoning related to finding volume with fractional side lengths as factors.</td><td>MA.K12.MTR.6.1: Reasonableness
Students will be working to solve real-world problems where they must find the total number of cube units that can be packed within a larger rectangular prism. Consider encouraging students to predict solutions before solving the problem.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In 8.2.3 Guided Practice, consider providing diagrams of blank rectangular prisms for students to use to help with questions 1 and 2. Students can mark up the sides of the prism with the given fractional units to assist them in finding the total number of cubes enclosed in a prism. Isometric dot paper is also a helpful support if students would like to draw the prisms from scratch.

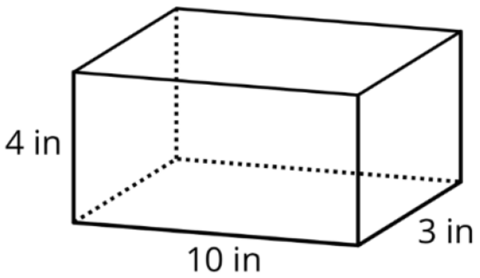
To foster kinesthetic reasoning about volume with fractional side lengths, provide unit cubes or similar math manipulatives for students to build prisms representing the problems in 8.2.4 Your Turn! or 8.2.5 Wrap-Up sections.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

8.2.1 Warm-Up: A Box of Cubes

Component Overview: *Students investigate volume by finding the total number of unit cubes inside a rectangular prism, recalling the previous lesson's activities. Students then determine the total number of cubes that can fit within the prism if the side lengths are increased or decreased.*

8.2.1 Warm-Up: A Box of Cubes

1. How many cubes with an edge length of 1 inch (in.) fill the box?



$V_{Box} = lwh = 4\ in \times 10\ in \times 3\ in = 120\ in^3$
$V_{Cube} = lwh = 1\ in \times 1\ in \times 1\ in = 1\ in^3$

$$\text{Number of cubes} = \frac{120\ in^3}{1\ in^3} = 120 \text{ cubes}$$

2. If the cubes had an edge length of 2 inches, explain if more or fewer cubes would be needed to fill the box.
Fewer cubes would be needed because the cubes would be bigger and fill more space.

3. If the cubes had an edge length of $\frac{1}{2}$ inch, explain if more or fewer cubes would be needed to fill the box.
More cubes would be needed to fill the box because the cubes would be smaller and take up less space.

Consider positioning students to discuss question prompts rather than writing their answers. This auditory entry point might engage learners and foster correct use of math vocabulary and problem-solving strategies. Use sentence stems to encourage thinking such as:

- *"I had to pay attention when problem-solving when…"*
- *"One detail that I have to remember a lot in math class is…"*

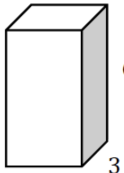
Consider prompting students to draw a sketch for both questions 2 and 3. This can help students visually make the connection between how the edge length affects how many cubes can fit into the box. It may be helpful to also have a physical manipulative to share whole group (comparing 1 unit to 2 units to $\frac{1}{2}$ unit).

8.2.2 Guided Instruction: Two Methods for Finding the Volume of a Right Rectangular Prism

Component Overview: *Students apply strategies to find the volume of right rectangular prisms. Question 1 involves a guided review on how to calculate the volume of the prism with fractional edges by packing cubes and by applying the volume formula ($V = lwh$). The second section leads students to discover that another formula ($V = Bh$) can be used to calculate the volume of a right rectangular prism.*

8.2.2 Guided Instruction: Two Methods for Finding the Volume of a Right Rectangular Prism

Alan calculated the volume of the rectangular prism.



1. Fill in the blanks to complete Alan's statement on how he calculated the volume.

To find the volume of this prism, I will fill the prism with cubes of side lengths $\frac{1}{3}$ in. Because $3\frac{2}{3} \div \frac{1}{3} =$ __11__, __11__ cubes will fit along the length and __11__ cubes will fit along the width. __121__ cubes will cover the base of the prism. The height of the prism can be used to determine how many layers of cubes are needed. It will take __20__ layers of cubes because $6\frac{2}{3} \div \frac{1}{3} =$ __20__. In total, it will take __2420__ of the $\frac{1}{3}$ in. cubes to fill the prism.

The volume of each small cube is $\frac{1}{27}$ in^3.

2. Multiply the number of cubes needed to fill the prism by the volume of each cube to find the volume of the prism.

$89\frac{17}{27}\ in^3$

Consider which questions students should use a calculator and which questions provide are good practice for fluency.

Is there another way to find the volume of this prism besides using these small cubes? Which strategy do you prefer?

8.2.2 Guided Instruction: Two Methods for Finding the Volume of a Right Rectangular Prism (continued)

3. Jai used the formula $V = l \times w \times h$ to find the volume of the prism. Complete Jai's work to find the volume of the prism.

$$V = l \times w \times h$$

$$V = 3\frac{2}{3} \times 3\frac{2}{3} \times 6\frac{2}{3}$$

$$V = 89\frac{17}{27}\ in^3$$

4. How are Alan and Jai's methods similar?
Answers will vary. Alan and Jai's methods were similar because they both multiplied the sides together.

The volume, V, of a rectangular prism is

$V = l \times w \times h$, where l represents the length, w represents the width, and h represents the height. This can also be written as $V = B \times h$, where B represents the area of the base and h represents the height.

5. Discuss with your partner how the formula $V = B \cdot h$ is related to the formula $V = l \cdot w \cdot h$.
The area of the base (B) is found by multiplying length times width.

6. Complete the statements based on your discussion.

The product of the length, l, and the width, w, is equal to

________*the area of the base*________ of the prism.

$B =$ __$l \times w$__

Both formulas can be used to find the volume of a right rectangular prism.

What is considered the "base" of a rectangular prism? If we rotate the prism, does the base change? Will the formula change?

Specifically have students make connections between $B = l \times w$ and the shape of the base in a rectangular prism. Since the base is a rectangle, the area formula used for B reflects the area of a rectangle ($A = l \times w$).

8.2.3 Guided Practice: Volume of a Right Rectangular Prism

Component Overview: *Students transition from teacher-led 8.2.2 Guided Instruction to student-led 8.2.3 Guided Practice. In this activity, students are prompted to both contemplate and calculate in order to have opportunity to make sense of information while finding solutions.*

8.2.3 Guided Practice: Volume of a Right Rectangular Prism

1. Diego correctly points out that 81 cubes with an edge length of $\frac{1}{3}$ centimeters (cm) are needed to fill a right rectangular prism that has length of 3 cm, width of 1 cm, and height of 1 cm.

   a. Explain or show how this is true.
   81 of the cubes with a volume of $\frac{1}{27}\,cm^3$ would have a combined volume of $3\,cm^3$ which matches the volume when multiplying $l \cdot w \cdot h$ of the right rectangular prism.

   b. What is the volume, in cubic centimeters, of the rectangular prism?
   $3\,cm^3$

2. Lin and Noah are packing small cubes into a cube with an edge length of $1\frac{1}{2}$ inches. Lin is using cubes with an edge length of $\frac{1}{2}$ inch, and Noah is using cubes with an edge length of $\frac{1}{4}$ inch.

   a. Explain who would need more cubes to fill the $1\frac{1}{2}$ in. cube.
   Noah would need more cubes because $\frac{1}{4}\,in$ cubes are smaller and fill less space than $\frac{1}{2}\,in$ cubes.

   b. Explain whether Lin and Noah will find the same volume of the $1\frac{1}{2}$ inch cube using their method.
   They will find the same volume using their own method because even though Noah will need more cubes, his cubes are smaller and will fill up the same amount of space.

Consider providing a diagram of a rectangular prism to manipulate for these exercises. This visual and kinesthetic support can help students better understand how to use given information to find volume of various prisms.

To extend student thinking and reasoning:

- *Does it matter which fractional unit cubes we use to find the volume?*
- *Could we use a different fraction unit cube for these examples?*

Number Talk
Question 2 provides students with the opportunity to apply different fractional unit cubes to find the volume of a prism. Consider encouraging partner pairs to calculate the volume by selecting a different fractional unit cube. They should confirm with one another that they have arrived at the same volume and see if they notice a pattern between the numbers they used.

8.2.3 Guided Practice: Volume of a Right Rectangular Prism (continued)

c. What is the volume of the cube with $1\frac{1}{2}$ inch edge lengths? Include units.
$3\frac{3}{8} in^3$

3. A toy company is packaging its toys to be shipped. Some small toys are placed inside a cube-shaped box with side lengths of 0.5 inches (in.). These smaller boxes are then packed into a shipping box with dimensions of 12 in. $\times$ 35 in. $\times$ 2.5 in.

   a. How many small toys can be packed into the larger box for shipping?
   $8{,}400$ *small toys*

   b. Use the number of toys that can be shipped in the box to determine the volume of the box. Include units.
   $1{,}050\ in^3$

For early finishers, consider an additional challenge by changing the size of the toys. Students could first make a guess for how many 1-inch or 2.5-inch-size cube boxes could fit and then calculate to test their hypotheses.

8.2.4 Your Turn!

Component Overview: *Students engage with real-world problems involving volume. Students should independently make sense of problems and apply methods of finding volume to practice in real-world contexts.*

8.2.4 Your Turn!

1. To give their animals essential minerals and nutrients, farmers and ranchers often have a block of salt, called a "salt lick" available for their animals. A rancher is ordering a box of cube-shaped salt licks. The edge lengths of each salt lick are $\frac{5}{12}$ foot (ft).

   a. Explain if the volume of one salt lick is greater than or less than 1 cubic foot (ft^3).
   The volume of one salt lick is less than 1 cubic foot because when you multiply the sides, which are fractions, your resulting product is smaller.

   b. A box that is $1\frac{1}{4}$ ft $\times$ $1\frac{2}{3}$ ft $\times$ $1\frac{5}{6}$ ft is used to ship the salt licks. How many cubes of salt lick can fit in the box?
   $1\frac{5}{12} \div \frac{5}{12} = 3$ *fit along this dimension of the box*
   $1\frac{2}{3} \div \frac{5}{12} = 4$ *fit along this dimension of the box*
   $1\frac{5}{6} \div \frac{5}{12} = 4.4$, *so* 4 *fit along this dimension of the box*
   $3 \times 4 \times 4 = 48$ *salt licks fit in the box*
   Some students may not recognize they can't cut salt licks to fit in the box, so may simply find the volume of each and divide to find 52.8 salt licks. This shows understanding of volume, but not the context.

2. A pool in the shape of a rectangular prism is being filled with water. The dimensions of the pool are 24 feet (ft) by 15 ft. If the height of the water in the pool is 1.3 ft, what is the volume of the water in cubic feet?
   $V = lwh = 24ft \times 15ft \times 1.3ft = 468ft^3$

Consider asking students what they know about salt licks. Allowing students to share something they have knowledge of that other students may not helps to build their confidence.

Review strategies to find the total number of salt cubes and strategies to determine the dimensions of the cube before moving on to the 8.2.5 Wrap-Up. Possible strategies include:

- *Applying the volume formula to the side lengths and then dividing by the volume of the salt cubes.*
- *Dividing each dimension by $\frac{5}{12}$ and then calculating the total number of cubes.*

Use this as an opportunity to clarify any lingering misconceptions.

Consider the entry point for question 1a and which students might benefit most from engaging with this question prompt. For additional complexity, have students attempt to provide counterexamples of when this might not be true.

What strategy for calculating the volume is most efficient for question 2? Why?

8.2.5 Wrap-Up: Ticket Out the Door

Component Overview: *Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' abilities to calculate the volume of a right rectangular prism and to find a total number of cubes encased within the prism.*

8.2.5 Wrap-Up: Ticket Out the Door

1. Joy keeps her modeling clay in a plastic box that has a length of 21 centimeters (cm), a width of 21 cm, and a height of 63 cm.

   a. What is the volume of the plastic box? Include units.

   $V_{\text{plastic box}} = 21\ cm \times 21\ cm \times 63\ cm = 27{,}783\ cm^3$

   b. If each piece of cube-shaped clay has a side length of 4.2 cm, how many pieces of clay can be stored in the box?

   $V_{\text{clay}} = lwh = 4.2cm \times 4.2cm \times 4.2cm = 74.088cm^3$

   $N_{\text{clay boxes}} = \dfrac{27{,}783\ cm^3}{74.088\ cm^3} = 375 \text{ boxes}$

For students who might struggle with math organization in multi-step problems, consider providing scaffolded steps for students to fill in to demonstrate their understanding. An example is below:

a. $V = lwh$

$V = \_\_\_\_ \times \_\_\_\_\_ \times \_\_\_\_\_$

$V = \_\_\_\_\_\_\_\_$

b. _________ *4.2 cm cubes fit in the length*

_________ *4.2 cm cubes fit in the width*

_________ *4.2 cm cubes fit in the height*

_________ *total 4.2 cm cubes*

8.3 Finding the Missing Dimension of a Rectangular Prism

Lesson Introduction

Benchmark	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.	Learning Target	• I can find the value of a missing dimension of a rectangular prism.

	Building On	Working Toward
Benchmark Coherence	MA.5.GR.3.3 Solve real-world problems involving the volume of right rectangular prisms, including problems with an unknown edge length, with whole-number edge lengths using a visual model or a formula. Write an equation with a variable for the unknown to represent the problem.	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

Essential Questions	• What formulas or strategies can you use to find the missing dimension of a rectangular prism? • How does knowing the volume of an object help find its dimensions?
Lesson Narrative	In this lesson, students build on their prior knowledge of solving real-world problems involving volume to find a missing dimension in a rectangular prism. Students collaborate with a partner to generate ideas on strategies to solve for missing dimensions, then solidify understanding through guided instruction and practice with given formulas for finding the missing dimensions. Students reflect on their understanding of finding volume and missing dimensions in the 8.3.6 Wrap-Up.

Component Summary	8.3.1 Warm-Up (~5min.)	Critical-thinking activity to find various solutions to area problem *(SE p. 16)*	8.3.4 Guided Practice (10 min.)	Guided practice finding missing dimension in context *(SE p. 19)*
	8.3.2 Collaboration (5 min.)	Collaboration on solving a real-world volume problem with a missing dimension *(SE p. 17)*	8.3.5 Your Turn! (5 min.)	Independent practice finding missing dimension *(SE p. 19)*
	8.3.3 Guided Instruction (15 min.)	Students explore a strategy to find missing dimensions of a prism *(SE p. 17)*	8.3.6 Wrap-Up (~5 min.)	Self-reflection on solving problems involving volume *(SE p. 20)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Face Additional Terms: • Volume • Dimension	• Four-Function calculator • (Optional) Isometric dot paper • (Optional) Unit cubes or math manipulatives • (Optional) Graphic organizer	• (Optional) Create a graphic organizer as explained in the Differentiate Instruction for students to use as a scaffold for recognizing and using given dimensions to find the missing values.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may forget to find the product of the two dimensions given to help find the missing dimension.
Students may not recognize that they need to divide the volume by product of the given dimensions to find the missing dimension.
Students may struggle to differentiate between the measurement of a one-dimensional length, the area of a two-dimensional face, and the volume of a three-dimensional figure.</td><td>Consider using a table to help students keep track of length, width, and height for each volume problem. This may help reinforce the need for all three units in solving problems.
Provide multiple opportunities to highlight the use of units in area and volume to reinforce the distinction between the two measurements.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect
8.3.3 Guided Instruction positions students to notice similarities between the structures of two formulas to find the missing dimension. Position students to recognize how both formulas work using the inverse operation (division) to find the missing factor and to connect this formula to previous knowledge on finding volume.</td><td>MA.K12.MTR.6.1: Reasonableness
Students will be working to solve real-world problems where they must find the measurement of a missing dimension in context. Consider encouraging students to predict solutions before solving the problems and use estimation to confirm the reasonableness of their solutions.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In 8.3.4 Guided Practice, consider providing isometric dot paper or math manipulatives for students to visualize or create the prisms in the real-world story problems to see the missing dimension.

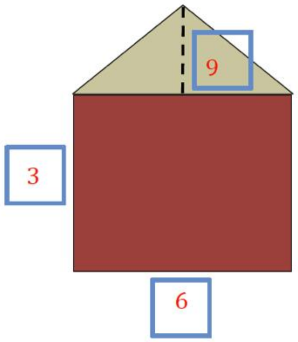
Consider providing graphic organizers to keep track of the dimensions or formulas with blank lines for students to fill in given values as scaffolded steps for students to fill in to assist in solving. Directly modeling and using a think aloud on how to substitute given values into the volume formula can also support auditory approaches to reasoning.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

8.3.1 Warm-Up: Area of a Wall

Component Overview: *Prompt students to find dimensions of the rectangle and triangle and their corresponding areas using the digits zero through nine once each at most, clarifying that there are multiple solutions.*

8.3.1 Warm-Up: Area of a Wall

1. Using the digits $0-9$, at most one time each, fill in the boxes to make a true statement.
   Answers will vary.



Area of  = 2 7

Area of  = 1 8

Consider supporting students who have difficulty getting started by reviewing how to find the area of a rectangle or triangle. For the rectangle portion, prompt students to consider what digits one through nine are factors of products that are also made up of digits one through nine. For the triangle, prompt students to recognize the length of the rectangle is the same measurement as the base of the triangle.

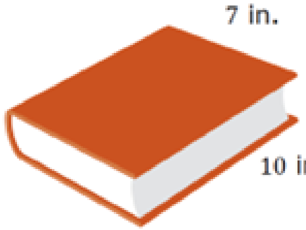
For an additional challenge, encourage students to find as many solutions as possible.

8.3.2 Collaboration: Missing Measurement

Component Overview: *Students collaborate on finding the missing measurement of a prism within the real-world context of a missing length of a textbook. Have students work in partners to calculate the volume of the old textbook by applying the volume formula, then dividing by the product of the given dimensions to find the missing dimension.*

8.3.2 Collaboration: Missing Measurement

Jolene works for a textbook company filling orders. She has discovered each textbook has a width of 7 inches (in.), a height of $2\frac{2}{3}$ in., and a length of 10 in.



The textbook company has decided to make the textbooks smaller by reducing the length. The new volume of the textbook will be 98 in^3.

1. Complete the table.

Old Textbook	New Textbook
$V = l \times w \times h$	$V = l \times w \times h$
$V = 10 \times 7 \times 2\frac{2}{3}$	$98\text{ in}^3 = l \times 7\text{ in.} \times 2\frac{2}{3}\text{ in.}$
$V = 186\frac{2}{3}\text{in}^3$	$98\text{ in}^3 = 5\frac{1}{4}\text{in.} \times 18\frac{2}{3}\text{ in}^2$

2. Work with your partner to determine the new length of the textbook. Explain your process.
   Answers will vary. I made an equation where length was the variable and I multiplied width times height and then divided 98 by my answer to get my new length. I multiplied $7 \cdot 2\frac{2}{3}$ which equals $18\frac{2}{3}$. Then I divided $98 \div 18\frac{2}{3}$ which equals $5\frac{1}{4}$.

Scaffold student thinking as they complete the table in question 1 by asking:

- *What values represent length, width, and height in the old textbook? How do you know?*
- *Which value did you write in the New Textbook column as the inches-squared value? Why?*
- *How can you use the volume formula and the old textbook information to help you find the new length of the textbook?*

Compare and Connect
Question 2 provides an important opportunity for students to compare the volume formula and make connections to how to use the inverse operation (division) to find the missing value. Position students to connect the equation for the new textbook to the volume formula and recognize that they are looking for a missing factor. Consider having students share their process with another partnership or to the class to address any misconceptions.

For struggling students, connect how to use division to find missing factors with simpler problems, like how we might find the missing factor in the equation $5 \times 2 \times a = 20$.

8.3.3 Guided Instruction: Finding a Missing Dimension

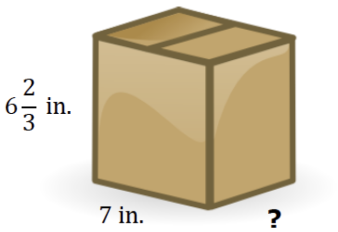
Component Overview: *Students are guided through two formulas or methods for finding a missing dimension on a rectangular prism when given the volume.*

8.3.3 Guided Instruction: Finding a Missing Dimension

Each flat surface of a prism is called a face.

When only two dimensions of a three-dimensional prism are given, the product of the two given dimensions represents the area of that face of the prism.

1. What is the area of the front face of this prism? Include units.



$46\frac{2}{3} in^2$

The volume of the prism is equal to the area of the face multiplied by the length of the missing dimension. The value of the missing dimension can then be calculated by dividing the volume by the area of the face.

$V = Area\ of\ the\ face\ \times measurement\ of\ the\ missing\ dimension$

$Measurement\ of\ the\ missing\ dimension = \dfrac{V}{Area\ of\ the\ face}$

For question 1, encourage math connections between vocabulary and volume concepts by asking:

- *What shapes are the faces of a rectangular prism?*
- *How do you find the area of a face of a rectangular prism?*
- *How can $V = lwh$ and $V = Bh$ both be used to find the volume of a rectangular prism and a missing dimension?*

Compare and Connect
Position students to compare and connect the two formulas for finding a missing dimension. Consider having students Think-Pair-Share with prompts like:

- *What is the same or different about these formulas?*
- *Can you use either formula to solve the problem?*
- *How are these formulas related to the formulas used to find volume: $V = lwh$ and $V = Bh$?*

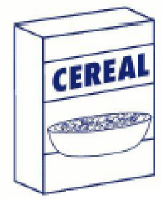
8.3.3 Guided Instruction: Finding a Missing Dimension (continued)

2. If the volume of the box is $233\frac{1}{3}\text{ in}^3$, what is the value of the missing dimension of the box?

$$V_{\text{box}} = \frac{700}{3}in^3 = 7\text{ in} \times w \times \frac{20}{3}\text{ in}$$

$$w = \frac{\frac{700}{3}in^3}{\frac{140}{3}in^2} = 5\text{ in}$$

3. Rafael was trying to find the missing dimension of the cereal box shown.



30 cm

20.3 cm

Volume = 1,218 cm^2

He made a mistake while solving for the width of the box, but he cannot find it. Inspect his work below.

Rafael's Work
20.3 cm $\times\ 30$ cm 60.9 cm^2 Area of face $60.9\overline{)12180}$ quotient 20 cm^3 Volume 1218 0

Write a note to Rafael on how to correct his error(s).

Raphael, it looks like you forgot to multiply by zero first when multiplying 20.3 times 30. This gave you the wrong area of the face, so your division was also incorrect. In addition, the units for your answer should be cm because you're only finding the width and dividing cm^3 by cm^2.

Consider modeling how to substitute values from question 1 into the volume formula when finding the missing dimension in question 2. It may be helpful to first substitute into the volume formula, showing the unknown variable as the measurement of the missing dimension, and then into the equation from question 2.

Prompt students to discuss the reasonableness of Rafael's answer while fostering a growth mindset by asking:

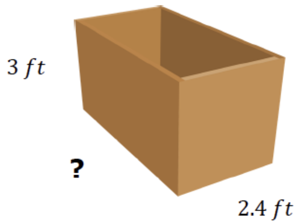
- *Notice Rafael's solution is the volume is 20 cm^3. Was he trying to find the volume or something else? Does the value he found make sense for the missing dimension?*
- *Look closely at Rafael's multiplication. What do you notice about the product and the factors? Is 60.9 cm^2 a reasonable answer to 20.3×30? Why or why not?*
- *How could you give kind, helpful, and specific feedback to Rafael to check his arithmetic? How could you help him consider a growth mindset when making mistakes?*

8.3.4 Guided Practice: Find the Missing Dimension

Component Overview: *Students practice finding the missing dimensions of rectangular prisms within two real-world contexts. Consider working through these problems as a class to address any remaining misconceptions.*

8.3.4 Guided Practice: Find the Missing Dimension

1. Sally broke her ruler before she finished measuring her toy chest. She knows the toy chest has a volume of $25.2\ ft^3$.



What is the length of the toy chest?

$l = 3.5\text{ ft}$

2. Damion works in construction and needs to cut a wooden beam to fill a cement hole. He knows that the beam has a width of 1.2 meters (m) and a length of 0.25 m. If Damion knows that the volume of the beam needs to be 1.5 cubic meters (m^3), what is the height?

$h = 5\text{ m}$

Encourage students to make predictions for measurements of the missing dimensions. Consider modeling through a think-aloud how to make predictions. For example, "The toy chest has a volume of $25.2\ ft^3$. Multiplying the width and height gives an area of $7.2\ ft^2$. The missing dimension should be around $3 ft$ since $7 \times 3 = 21$."

Provide a labeled image or create a rough drawing of a wooden beam to help students recognize the shape as a rectangular prism. Then scaffold student thinking for question 2 when solving as a class by asking:

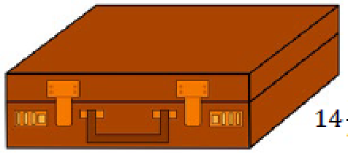
- *What formula should we use to find the missing dimension? Why?*
- *How do we calculate the area of the face?*
- *If we know the area of the face and volume, what operation is needed to find the measurement of the missing dimension?*
- *Is our solution reasonable? How can we use estimation to find out?*

8.3.5 Your Turn!

Component Overview: *Students independently practice finding missing dimensions given two dimensions and the volume.*

8.3.5 Your Turn!

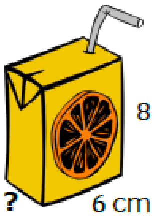
1. Mr. Rodrigo knows that his briefcase has a volume of $1{,}068\frac{3}{4}$ cubic centimeters (cm^3). He labeled the dimensions of the briefcase that he knew.



What is the height of the briefcase?

$h = 5$ cm

2. Victor works in a factory that makes juice boxes. He knows that the juice boxes have a height of 8 centimeters (cm), a length of 6 cm, and a volume of 192 cubic centimeters (cm^3).



What is the width of the juice box?

$w = 4$ cm

Consider providing a graphic organizer that prompts students to write the given values into the volume formulas for finding a missing dimension from 8.3.4 Guided Practice question 2. This can help students recognize what values are missing and how to solve. The organizer could be laminated, and students could use a dry-erase marker to write, erase, and rewrite given dimensions and values in the formula for all problems.

Consider providing additional challenge problems for students who may finish early or are ready for higher cognitive demand.

- *How could you change the dimensions of the juice box so that it could hold double the amount of juice? Half?*
- *A swimming pool can hold $2250 ft^3$ of water. The length and width are in a ratio of $1:2$. The height and length are in a ratio of $1:3$. Find the dimensions of the swimming pool.*

8.3.6 Wrap-Up: Reflection

Component Overview: *Students reflect on and rate their understanding from this and previous lessons on how to find volume of a rectangular prism and how to find a missing dimension of a rectangular prism. Consider alternative methods for students to share their thinking and explanations for the ratings.*

8.3.6 Wrap-Up: Reflection

1. Rate your strength on the following areas using the scale.
   Answers will vary.

   1 – still learning
   2 – making progress
   3 – got it

Skill	Rating
Finding the volume of a rectangular prism.	3
Find a missing dimension of a rectangular prism.	2

2. Write an explanation of why you rated yourself the way you did.
   Answers will vary. I can find the volume by multiplying length times width times height. Sometimes I get confused solving for a missing dimension when fractions are involved.

For students who might struggle representing their thoughts through writing, consider alternative ways for students to provide their reflection reasoning. Students could choose to record themselves explaining their reasoning verbally, drawing and labeling, or providing an example.

8.4 Nets of Rectangular Prisms and Pyramids

Lesson Introduction

Benchmark	MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's nets.	Learning Targets	• I can draw the net of a rectangular prism or rectangular pyramid. • I can relate a net to its three-dimensional shape. • I can relate a rectangular prism or rectangular pyramid to its net.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

Essential Questions	• What is a net? • What are the similarities or differences between a net of a three-dimensional figure and a composite figure? • What are some differences between the nets of rectangular prisms and the nets of rectangular pyramids?
Lesson Narrative	This lesson begins with a review of the previous lesson finding a missing dimension when given volume. The lesson begins with a review on composite figures and relates the composite figures to the net of a three-dimensional figure. Students explore how nets represent the two-dimensional faces of the three-dimensional figures and collaboratively cut out, fold, and create prisms to match three-dimensional solids to the nets that they form when folded. The lesson wraps up with students' self-reflections on their learning about nets.

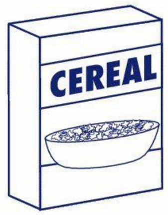
Component Summary	8.4.1 Warm-Up (~5 min.)	Review finding missing dimension of prism when given volume *(SE p. 21)*	8.4.3 Exploration (5-10 min.)	How three-dimensional figures relate to given nets *(SE p. 22)*
	8.4.2 Refresher (10min.)	Review how to find the total area of composite figures *(SE p. 21)*	8.4.4 Collaboration (10-15 min.)	Draw and label nets for the given three-dimensional figures *(SE p. 23)*
	8.4.5 Wrap-Up (~5 min.)	Reflect on the most difficult part of the lesson *(SE p. 24)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Net <u>Additional Terms:</u> • Face • Cube	• Nets for 8.4.3 Exploration • Scissors • (Optional) Tape for constructing nets • Four-function calculator	• Print and cut out nets for 8.4.3 Exploration • (Optional) Gather real-world examples of nets and models of three-dimensional solids.

Teacher Tips

Common Misconceptions	Things to Consider
• Students may struggle with labeling all the sides or faces in a net. • Students may struggle recognizing or finding missing side lengths. • Students may struggle with visualizing the three-dimensional solid that the nets create. • Students may struggle with connecting different nets to the same three-dimensional figure.	• Students should be monitored closely during the 8.4.3 Exploration to be sure that they cut and fold the nets to correctly form the intended solid. Having examples for students to refer to may be helpful. • Teacher scaffolding through the 8.4.2 Refresher may be beneficial to help students build connections between two-dimensional composite figures and the prisms they represent. • Consider gathering or having students recognize real-world examples of nets (e.g., wrapping a box in gift paper or unfolding a shoe box).
Literacy and Language Routines	**Mathematical Thinking and Reasoning (MTR) Standard**
Compare and Connect Like the previous lesson, this lesson provides important opportunities for students to compare and connect concepts related to representing three-dimensional figures and their characteristics. Consider fostering discussion or using a Venn diagram to organize and record student thinking during 8.4.3 Exploration as students compare the idea of a composite figure and a net of a three-dimensional figure.	**MA.K12.MTR.2.1: Multiple Representations** As students connect two-dimensional nets to three-dimensional prisms or pyramids, they are deepening their understanding of how to represent shapes in multiple ways. Clarifications within this standard emphasize connections between concept and representation. Utilize the progression of the lesson to highlight how nets create three-dimensional solids, including by physically cutting out and building prisms from printed nets.

Consider extending the lesson by having students unfold and refold real-life examples of three-dimensional figures, such as shoe boxes, after completing 8.4.3 Exploration. Allowing students physical models of the shapes may strengthen the connection between the nets and the solids for learners.



Lesson Notes:

8.4.1 Warm-Up: Bell Work

Component Overview: *Students are given the volume, length, and depth of a real-world three-dimensional shape and find the missing width. Have students work individually and consider using responses as an additional data point or formative assessment of learning targets from Lesson 3.*

8.4.1 Warm-Up: Bell Work

1. A rectangular ball pit has a volume of 28 cubic meters (m^3). Find the width of the ball pit if the length is $5\frac{5}{6}$ m and the depth is $\frac{3}{4}$ m.

$$28\ m^3 = \frac{35}{6}\text{m} \times w \times \frac{3}{4}\text{m}$$

$$28\ m^3 = \frac{105}{24}m^2 \times w$$

$$w = \frac{28\ m^3}{\frac{105}{24}m^2} = \frac{32}{5}\text{m}$$

$$w = 6\frac{2}{5}\text{m}$$

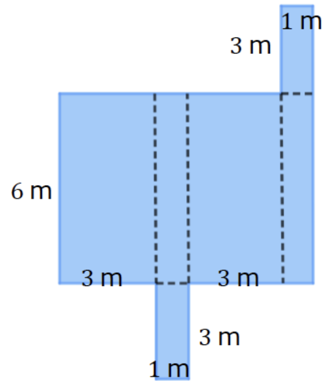
Pictures or drawn models of the scenario can provide clarity and strengthen understanding. Encourage the students to draw a labeled picture of the figure described in question 1 prior to solving the problem and to label the length, depth, volume, and eventually the width as they work.

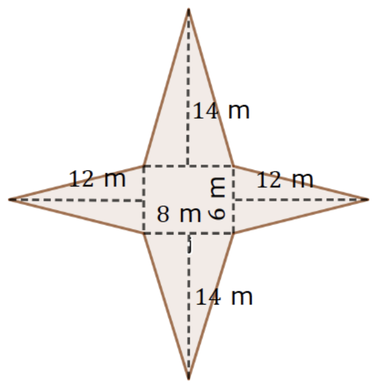
8.4.2 Refresher: Area of Composite Figures

Component Overview: *This section reviews how to find the total area of composite figures. Students find missing dimensions and determine the combined area of various rectangles and triangles, and though it is not yet connected, students are finding surface area of a rectangular prism and pyramid. As students work in partners, encourage students to label all missing dimensions prior to finding the area and have students use formulas to find the area. Consider working with a small group of students to scaffold review of area for any students who need additional support.*

8.4.2 Refresher: Area of Composite Figures

1. Work with your partner to find the area of each composite figure. All measurements are in meters (m).

<table>
<tr><th>Figure</th><th>Area</th></tr>
<tr><td>All angles are right angles.


</td><td>$54\ m^2$</td></tr>
<tr><td>All dashed lines intersect at right angles.


</td><td>$232\ m^2$</td></tr>
</table>

To support connecting these composite figures to the nets of rectangular and triangular prisms, pose these questions:

- *What shapes are included when finding the total area of the first figure?*
- *What shapes are included when finding the total area of the second figure?*

Compare and Connect
Compare and contrast the similarities and differences in finding the area of each of the composite figures. Students may point out that the first figure is made up of six shapes and the second figure is made up of fives shapes. Students may also respond that the second shape appears to have a shape in the center that is a different shape than the shapes surrounding it.

8.4.3 Exploration: Creating Three-Dimensional Figures

Component Overview: *Students discover which nets represent which three-dimensional figure: a rectangular prism and a pyramid. Students also make connections to composite figures and nets in questions 3 and 4. Consider explaining question 1 together as a class, then prompting students to work in partners or small groups as they complete the table in question 2. Be sure to provide enough copies of nets and scissors to each group and encourage students to fold and connect the edges of the net to see the prism or pyramid come to life through folding.*

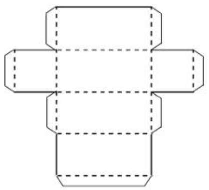
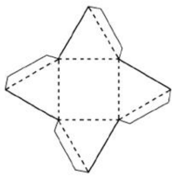
8.4.3 Exploration: Creating Three-Dimensional Figures

Both composite figures in the previous component are nets. A net is formed when the faces of a three-dimensional figure are positioned by arranging the faces to create a two-dimensional composite figure.

A ***net*** is a two-dimensional diagram that can be folded or made into a three-dimensional figure.

1. Your teacher will provide you with copies of some nets. Cut the nets along the solid line. Fold the nets along the dotted line and use the tabs to glue the figure together.

2. Complete the table to describe the three-dimensional figure formed.

Net	Three-Dimensional Figure Formed
	*Rectangular Prism*
	*Pyramid*

Allow students to answer these questions as they explore to strengthen the connection from composite figures to the nets:

- *What is similar about a net of a three-dimensional figure and a composite figure?*
- *What are some real-world examples of nets?*

8.4.3 Exploration: Creating Three-Dimensional Figures (continued)

3. Look back at the two composite figures from the refresher. Predict what three-dimensional figure is made from cutting out and folding up the sides of those figures.
   Rectangular Prism, Pyramid

4. With your partner, discuss how the area of a two-dimensional composite figure is related to the area of the faces of the three-dimensional figure. Summarize your discussion.
   The area of a 2-dimensional composite shape will help you find the area of the outside of a 3-dimensional shape.

Compare and Connect
At the end of the Exploration, regroup students as a class to discuss responses to questions 3 and 4. Position students to make connections between composite figures as the faces of three-dimensional figures. Use real-world examples, like wrapping a box in paper, to solidify that the area of the two-dimensional shape is the same as the area of the outside of the three-dimensional shape. Ask:

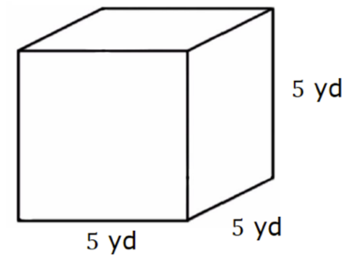
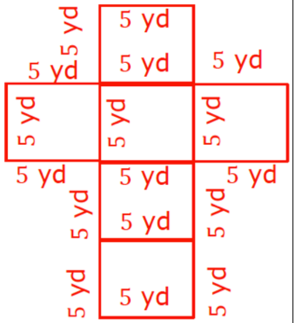
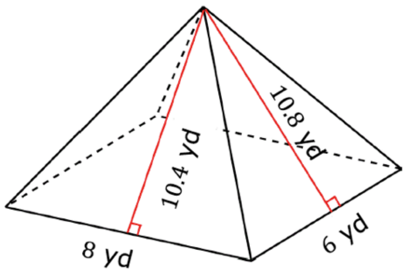
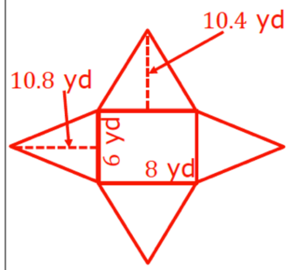
- *What shapes are the faces of a pyramid?*
- *What shapes are the faces of a prism?*

8.4.4 Collaboration: Determining Nets

Component Overview: *Students collaborate by sketching and labeling nets for two three-dimensional figures: a cube and a rectangular pyramid. Students build on previous knowledge by identifying dimensions of the figures. Before students begin the collaboration, discuss ideas for the individual two-dimensional shapes that make up the faces of each three-dimensional figure. Encourage students to be sure that the individual shapes, or faces, are all connected so that none of the shapes are drawn by themselves.*

8.4.4 Collaboration: Determining Nets

1. With your partner, sketch the net or sketch the individual faces of the three-dimensional figure. Label each side of the net with the appropriate measurement in yards (yd).

<table>
<tr><th>Three-Dimensional Figure</th><th>Net</th></tr>
<tr><td>
</td><td>
</td></tr>
<tr><td>
</td><td>
</td></tr>
</table>

Prior to starting the 8.4.4 Collaboration, have students match the nets made from 8.4.3 Exploration with the solids in the table. Though the dimensions are different, students may benefit from visually representing the same shape as a two-dimensional collection of shapes as they draw the nets in question 1. Also consider providing additional examples of drawings of the shapes or physical models and providing different nets that match to the same three-dimensional figure.

Reinforce matching dimensions on different kinds of rectangular prisms by asking the students the following questions:

- *How were you able to calculate all the missing sides for the prism and pyramid in the table?*
- *How can you tell the figure in the table is a cube? What shape are the faces of a cube?*
- *How are cubes and rectangular prisms the same?*

8.4.5 Wrap-Up: Reflection

Component Overview: *Students reflect on the lesson to determine the most difficult or confusing point on the lesson about nets. Consider collecting these responses to address misconceptions or reinforce challenging concepts at the close of the lesson.*

8.4.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on nets?

Answers will vary. The muddiest thing about today's lesson for me was trying to visualize what the sides of the 3-dimensional shape looked like. The muddiest thing about today's lesson for me was trying to visualize what a shape on paper looks like folded together.

Consider asking students to complete a table where they will identify the clearest point of the lesson and one unclear point of the lesson as well as the muddiest point of the lesson. By encouraging students to think about a range of understanding, students may avoid responding that the entire lesson was "muddy."

Consider prompting students to provide evidence for their responses. Providing sentence stems for them to complete may assist with communicating their thoughts clearly, such as:

- *"The muddiest point was_____ because_________."*
- *"I found the point _______ confusing because I did not understand ___________________________."*

8.5 Surface Area of Rectangular Prisms and Pyramids

Lesson Introduction

Benchmark	MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.	Learning Targets	• I can find the surface area of a rectangular prism using a net or the figure. • I can find the surface area of a rectangular pyramid using a net or the figure.

Benchmark Coherence	Building On	Working Toward
	N/A	MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

Essential Questions	• What does the net of a rectangular prism look like? A rectangular pyramid? • What is surface area? • What strategies can you use to find the surface area of a rectangular prism or pyramid? • How can a net be used to find the surface area of a rectangular prism or pyramid?
Lesson Narrative	In this lesson, students are introduced to finding the surface area of nets. Through guided instruction, students find the surface area of a rectangular prism and pyramid using a net by first labeling given dimensions and finding the area of each face. By the end of the lesson, students show their learning by identifying all dimensions, labeling nets accurately, and finding the surface area of real-world examples of rectangular prisms and pyramids.

Component Summary				
	8.5.1 Warm-Up (~5 min.)	Which One Doesn't Belong? activity with patterns *(SE p. 25)*	8.5.4 Guided Instruction (10-15 min.)	Introduction to pyramids and finding the surface area using nets *(SE p. 28)*
	8.5.2 Guided Instruction (10-15 min.)	Guided instruction on finding the surface area of rectangular prisms using nets *(SE p. 25)*	8.5.5 Collaboration (~5 min.)	Collaborate to find the surface area of a pyramid *(SE p. 29)*
	8.5.3 Collaboration (~10 min.)	Compare student work *(SE p. 27)*	8.5.6 Your Turn! (~5 min)	Individual practice determining surface area of rectangular prisms and pyramids within real-world contexts *(SE p. 30)*
	8.5.7 Wrap-Up (~5 min.)	Assessment on using a net to label dimensions and find surface area of a rectangular prism *(SE p. 31)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Surface area • Slant height <u>Additional Terms:</u> • Net • Face	• (Optional) Four-function calculator • (Optional) Graph paper to draw and cut out nets or preprinted nets	• Examples of nets of square pyramids and of rectangular pyramids may be beneficial.

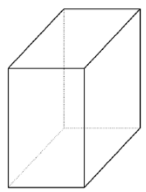
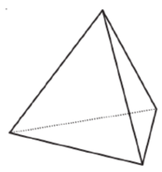
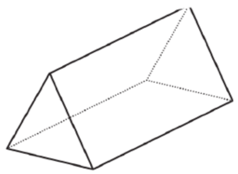
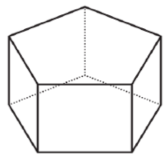
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may miscalculate the missing dimensions.
Students may struggle with correctly labeling the dimensions on the nets.
Students may forget to find the sum of all the areas of the faces or miss one of the faces.
Students may confuse the height and the slant height.
Students may assume that all the triangle faces in a pyramid have the same dimensions or the same area.</td><td>Finding the surface area of three-dimensional figures is a lengthy process for students to execute. Provide different methods to finding the surface area and checkpoints throughout problem-solving to assist students with resilience. Monitor student problem-solving closely for misunderstanding.
Reinforce, as nets and as solids, that all rectangular prisms have exactly six faces and all rectangular pyramids have exactly five faces. Their combined surface area should include the sum of six faces and five faces, respectively.
Determine which problems, if any, you think your students would benefit from using a calculator on. When determining which problems, consider the difficulty of the problem and if that difficulty would be a barrier to student learning.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Collect and Display
Consider implementing Collect and Display routines after completing the 8.5.3 Guided Practice. While the students are completing the question, collect comments they make during their discussions about the strategies. Display these comments and others to brainstorm what these comments could be referring to when their peers made them.</td><td>MA.K12.MTR.7.1: Real World Contexts
This standard emphasizes experiencing math through real-world examples. In this lesson, students explore various authentic examples of rectangular prisms and work with nets to find and label dimensions and find the total surface area in real-world contexts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider providing graph paper, rulers, markers, scissors, and tape for students to recreate the nets in 8.5.3 Collaboration and 8.5.6 Your Turn!. By providing these hands-on materials and visuals, students can create prisms to scale and physically label each face as they measure and find the area, additionally helping students link the characteristics of two-dimensional nets to the characteristics of three-dimensional rectangular prisms.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

8.5.1 Warm-Up: Which One Doesn’t Belong?

Component Overview: *Students are presented with four patterns. Students must determine which pattern does not belong with the other three and must justify their selection.*

8.5.1 Warm-Up: Which One Doesn’t Belong?

1. Four figures are shown.

A	B
	
C	**D**
	

a. Which one doesn’t belong?
Answers will vary.

b. Justify your choice using mathematics.
A is the only one where all of the faces are rectangles; A is the only rectangular prism.

B is the only one where all the faces are triangles; or the only one that doesn’t include rectangular faces; or the only one that isn’t a prism; or the only one that is a pyramid.

C is the only one that include rectangular and triangular faces.

D is the only one with faces that aren’t rectangles or triangles; is the only one with pentagonal faces.

Consider a kinesthetic approach to eliciting student responses to this activity. Post a picture of each pattern, one per corner, and ask the students to stand in the corner with the picture that does not belong. Ask students to share their justifications first with their group, then to other groups in other corners. Allow students to walk between corners as an added kinesthetic approach to clarify and justify thinking.

8.5.2 Guided Instruction: Surface Area of a Rectangular Prism

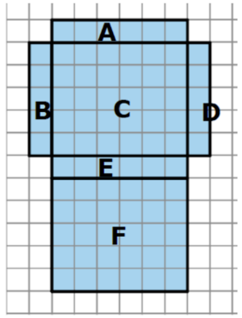
Component Overview: *Students are introduced to finding the surface area of two rectangular prisms using nets. Students are given a net on a grid and must find all the dimensions, the area of each face, and the surface area. Students are also given a real-world example of a rectangular prism and must draw the net, label it, and find the surface area. Through guided instruction and discussion, position students to identify the shape of each face and to find the area of each shape.*

8.5.2 Guided Instruction: Surface Area of a Rectangular Prism

To find the surface area of a three-dimensional object, use the nets of a shape to find the area of each face.

Surface area is the total area of the two-dimensional surfaces that make up a three-dimensional object

1. A net of a rectangular prism is shown below with each face labeled.
   a. Complete the table to find the area of each face. Each unit of the grid is 1 centimeter (cm).



Face	Dimensions	Area
A	$1\ cm\ x\ 6\ cm$	$6\ cm^2$
B	$1\ cm\ x\ 5\ cm$	$5\ cm^2$
C	$5\ cm\ x\ 6\ cm$	$30\ cm^2$
D	$1\ cm\ x\ 5\ cm$	$5\ cm^2$
E	$1\ cm\ x\ 6\ cm$	$6\ cm^2$
F	$5\ cm\ x\ 6\ cm$	$30\ cm^2$

   b. How can the net and the area of the six faces be used to find the surface area of the prism?
   You can add up the areas of all the sides to find the surface area.

   c. What is the surface area of this rectangular prism?
   $82\ cm^2$

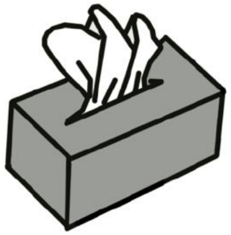
When calculating surface area, find the sum of the areas of each face of the figure. Use the figure's net to label the dimensions needed to calculate the area of each face.

For question 1, elicit student thinking on key elements about the figure before completing the table by asking:
- *How can we find the dimensions of each shape?*
- *How can we verify that the net of the prism is made up of rectangles and not squares?*
- *Since the net is on a grid, what is another method we could use to find the surface area instead of adding all the areas?*

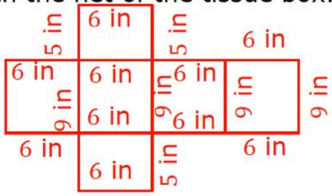
Before reading the blue box explaining how to calculate surface area aloud, have students read it individually to themselves, circling any words they do not recognize or are unfamiliar with. Then define any needed words and read the description aloud. This auditory approach may be especially helpful for ELL students by frontloading key vocabulary content that is needed to access procedural knowledge on finding surface area using a net.

8.5.2 Guided Instruction: Surface Area of a Rectangular Prism (continued)

2. A tissue box is 9 inches (in.) long, 5 in. wide, and 6 in. tall.



a. Sketch the net of the tissue box.



b. Label the dimensions of each face on the net.

c. Find the area of each face. Include units.

$45\ in^2$
$45\ in^2$
$54\ in^2$
$54\ in^2$
$30\ in^2$
$30\ in^2$

d. Add the area of the faces to find the surface area of the tissue box. Include units.

$258\ in^2$

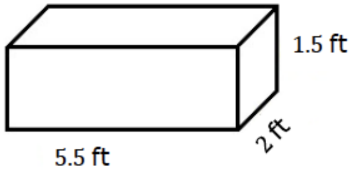
Collect and Display
After students complete question 2, prompt students to consider a different arrangement of the faces of the tissue box. Consider displaying labeled nets that are not identical to each other but still use the correct dimensions, then discuss how they are all correct. Students may point out that all the nets have six shapes. Students may also share that when folded, the nets form a rectangular prism. Point out that the arrangement of the six shapes could be different, but when folded, they still create the same rectangular prism.

8.5.3 Collaboration: Analyzing Student Work

Component Overview: *Students work in partners to solve real-world problems related to the surface area of rectangular prisms, including labeling a net's dimensions and finding the surface area by first finding the area of each face. Consider regrouping and sharing ideas at the end of 8.5.3 Collaboration section as students compare two different strategies used to find surface area.*

8.5.3 Collaboration: Analyzing Student Work

1. Jillian and Victor are finding the surface area, in square feet (ft^2), of the rectangular prism shown and each got the same answer; however, they solved the problem differently.



Jillian's Work	Victor's Work
Side 1:	2 sides:
$5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$	$2(5.5 \text{ ft} \times 1.5 \text{ ft}) = 16.5 \text{ ft}^2$
Side 2:	2 sides:
$5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$	$2(5.5 \text{ ft} \times 2 \text{ ft}) = 22 \text{ ft}^2$
Side 3:	2 sides:
$5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$	$2(2 \text{ ft} \times 1.5 \text{ ft}) = 6 \text{ ft}^2$
Side 4:	
$5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$	Total Surface Area: $44.5\ ft^2$
Side 5:	
$2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$	
Side 6:	
$2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$	
Total Surface Area: 44.5 ft^2	

Discuss with your partner each student's strategy and how they are similar or different. Draw arrows to show connections between each student's work.

Answers will vary.

Jillian found the area of each side then add them all together. Victor knew that 2 sides that were parallel to each other would have the same measurements therefore, he multiplied the 3 different side areas by 2 and then added. They both added the side areas.

Extend and connect geometric concepts by prompting students to identify congruent shapes. Ask students how to determine if a shape is similar or congruent, then connect this concept to nets by identifying each pair of congruent sides and justify their reasoning. Students may respond that the top and the bottom are congruent because the dimensions are identical.

Collect and Display

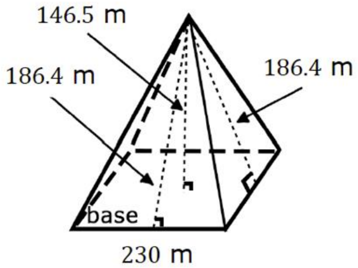
After students have discussed each student's strategy and how they are all similar and different, consider having students write which strategy they prefer on a sticky note. Display these sticky notes on a shared classroom space and discuss which strategy is more popular. Point out that both strategies work, but the process and approach differ. Encourage students to try the strategy they did not pick on a future problem and return to this discussion after question 1 in 8.5.6 Your Turn!.

8.5.4 Guided Instruction: Surface Area of a Rectangular Pyramid

Component Overview: *Students are introduced to pyramids using Egyptian pyramids and focus on finding the surface area when given a 3D shape and/or net. Instruction should include the difference between the height and the slant height. Students label the net, find the area of each face, and find the total surface area, then select correct options to complete statements on finding the surface area of pyramids.*

8.5.4 Guided Instruction: Surface Area of a Rectangular Pyramid

1. Around 2500 B.C. the Egyptians began building pyramids with intricate hallways and layouts to house the tombs of pharaohs and their families. One of the most famous is the Pyramid of Khufu, seen below. The Egyptians designed the pyramid to have a square base with approximate side lengths of 230 meters, a height of 146.5 meters, and a slant height of 186.4 meters.



Looking at the model of the pyramid, why is there a measurement called height and a measurement called slant height?
The height is straight up the middle of the pyramid to the highest point. The slant height is up the side of the pyramid to the top because the side is slanted.

The slant height of the pyramid represents the height of each triangular face.

When finding the surface area of a pyramid, use the **slant height** to find the area of the faces.

While looking at the model of the pyramid in question 1, prompt students to consider the key elements distinguishing the height from the slant height by asking:

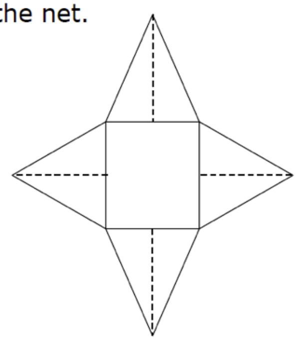
- *Where in the diagram are the height of the pyramid and the slant heights of the faces located?*
- *Why does the diagram include more than one slant height and only one height?*
- *What does the square box at the bottom of the height indicate?*

Consider using a hands-on demonstration to help students differentiate between slant height and pyramid height. Use a physical net of a pyramid and ask students to identify and point at the slant height of each triangular face on the physical net and on the drawn net in question 1. Then fold the net together and ask students to identify the pyramid height on the physical net and on the drawn pyramid.

Students may find it easier to understand slant height by comparing it to the stairs needed to walk up the side of the pyramid.

8.5.4 Guided Instruction: Surface Area of a Rectangular Pyramid (continued)

2. Using the information about the pyramid, complete the table to determine the surface area of the pyramid. Begin by labeling the sides of the net.



Side	Formula	Area
Rectangular base	$A = l \cdot w$	$A = 230 \cdot 230$ $A = 52{,}900\ m^2$
Triangular Face 1	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21{,}436\ m^2$
Triangular Face 2	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21{,}436\ m^2$
Triangular Face 3	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21{,}436\ m^2$
Triangular Face 4	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21{,}436\ m^2$
		Surface Area: $138{,}644\ m^2$

3. Complete the following statement.

To find the surface area of a rectangular pyramid you need the sum of the
○ height
○ area
of the base plus the area of
○ one of
○ all
the other faces.

Encourage students to analyze the net provided prior to completing the table. Students may point out that the center shape is a square because all the side lengths are the same. Students may also note that all the faces attached to the square are triangles. Students may also notice that squares are special kinds of rectangles.

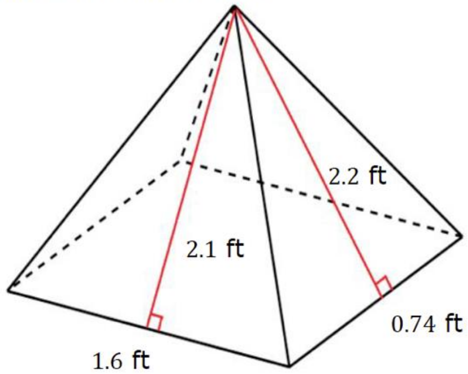
Also compare the net of a pyramid to the net of a rectangular prism and look for similarities and differences, like the number and shape of the faces.

8.5.5 Collaboration: Surface Area of a Rectangular Pyramid

Component Overview: *Students work with a partner to find the total surface area of a labeled diagram of a real-world example of a rectangular pyramid.*

8.5.5 Collaboration: Surface Area of a Rectangular Pyramid

Papier-mache is a type of art that uses paper and glue. Jose is using the art of papier-mache to create a rectangular pyramid for his art project. To create his project, Jose needs to determine the amount of area that will be covered in newspaper. Below is the drawing Jose created to find the surface area.



1. Work with your partner to find the surface area of the pyramid.

$$A_{rectangular\ face} = 1.6 \times 0.74 = 1.184\ ft^2$$

$$A_{triangular\ face\ 1} = \frac{0.74 \times 2.2}{2} = 0.814\ ft^2$$

$$A_{triangular\ face\ 2} = \frac{1.6 \times 2.1}{2} = 1.68\ ft^2$$

$$A_{triangular\ face\ 3} = \frac{0.74 \times 2.2}{2} = 0.814\ ft^2$$

$$A_{triangular\ face\ 4} = \frac{1.6 \times 2.1}{2} = 1.68\ ft^2$$

$$Total\ Surface\ Area = 1.184 + 0.814 + 1.68 + 0.814 + 1.68 = 6.172 ft^2$$

To scaffold students through the process to find the surface area, pose these questions:

- *What are the five shapes that make up this pyramid?*
- *Which of these shapes are congruent?*
- *What are the dimensions of the base and of each face?*
- *How can the area of the shapes be found?*
- *Once the areas are found, how can the total surface area be found?*

Finding the surface area of three-dimensional figures is a process that involves many steps for students to execute. Encourage students to draw labeled pictures as they go through the steps to find the surface area. Also encourage students to use a table, like the table included in the 8.5.4 Guided Instruction, for them to organize their calculations.

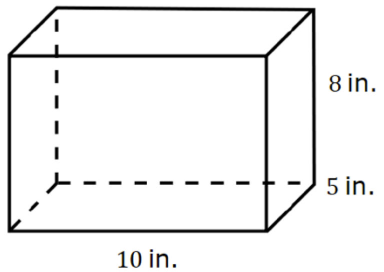
Consider an interdisciplinary lesson with an art teacher using papier-mâché to conceptualize and represent surface area. Have students create their own version of a net of a pyramid using cardstock paper, then measure the surface area to determine how much newspaper they would need to cover the pyramid with three layers. If time allows, let students create these pyramids and papier-mâché the outside surface area to display in the classroom.

8.5.6 Your Turn!

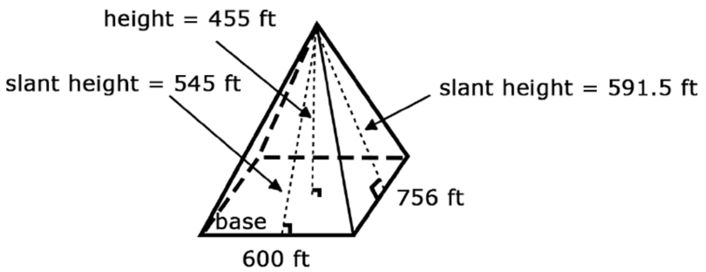
Component Overview: *An image of a rectangular prism is given, and students practice finding the surface area. Students are also given a real-world scenario involving a rectangular prism and must find the surface area. While some students work individually to practice these skills, consider working with a group of students who may need additional support identifying the shape of each face, labeling the dimensions, and finding the area of each shape. Allow students to draw nets that are not identical to each other.*

8.5.6 Your Turn!

1. Darsey wants to build trinket boxes made of wood to sell at the local craft festival. A drawing of her box is shown. How much wood does Darsey need to buy for each trinket box?
   $340\ in^2$



2. Find the surface area of a rectangular pyramid that has a rectangular base with a side length of 600 feet (ft), which is the base of the triangle with a slant height of 545 ft, and a side length of 756 ft, which is the base of the triangle with a slant height of 591.5 ft.



$Surface\ Area = 1{,}227{,}774\ ft^2$

Consider prompting students to draw a net for question 1, then recording the area of each shape inside of the shape on their drawings of the net. By labeling the area of each face, students practice a visual strategy that will assist them with finding the surface area.

For question 1, consider challenging students by presenting the same labeled rectangular prism but turned differently, such as one prism shown horizontally and the other prism shown vertically. Ask half of the class to solve the problem using the horizontal figure and the other half of the class to solve the problem using the vertical figure. Providing the same image in two different views reinforces that surface area can be found regardless of the positioning of the rectangular prism.

Prior to completing question 2, prompt students to notice specific characteristics of the pyramid. Students may recognize that the height of the pyramid is not necessary to calculate the surface area. Students may also see that since two different slant heights are labeled, all the triangular faces are not identical.

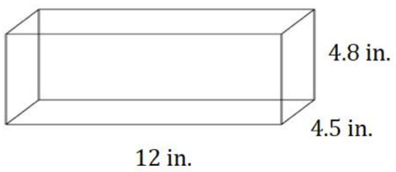
8.5.7 Wrap-Up: Ticket Out the Door

Component Overview: *Students individually solve two problems by labeling a net and finding surface area within a real-world context. Consider using student responses as a formative assessment of this lesson's learning targets.*

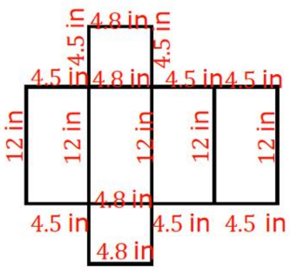
8.5.7 Wrap-Up: Ticket Out the Door

Ted is preparing a holiday display for the post office. He takes an unfolded box and uses it to determine the amount of wrapping paper needed to wrap the box.

The shipping box Ted uses for the display is shown.



1. Write the measurements on the net of the box.



2. Use the net to find the surface area of the shipping box. Include units.
   $266.4\ in^2$

Post these questions to reinforce the essential questions for early finishers or all students based on available time:

- *How do you know what the shapes of the faces are?*
- *Which faces have the exact same area? How can you tell? How many faces have the same area?*
- *How can you use the net to find the total area of all the surfaces of this prism?*

8.6 Finding Surface Area and Volume

Lesson Introduction

<table>
<tr><td rowspan="1">Benchmarks</td><td>MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure’s net.
MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula.</td><td>Learning Targets</td><td colspan="2">• I can find the surface area of right rectangular prisms and right rectangular pyramids.
• I can find the volume of right rectangular prisms.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td>Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td>MA.5.GR.3.3 Solve real-world problems involving the volume of right rectangular prisms, including problems with an unknown edge length, with whole-number edge lengths using a visual model or a formula. Write an equation with a variable for the unknown to represent the problem.</td><td colspan="3">MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.
MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.</td></tr>
<tr><td>Essential Questions</td><td colspan="4">• What is different between surface area and volume?
• How do you find the volume and surface area of a rectangular prism?
• How do you find the surface area of a rectangular pyramid?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">This lesson reviews skills and concepts from this unit including finding volume and surface area of rectangular prisms and pyramids and finding the missing dimension when given the volume. Through collaboration and guided practice, students strengthen previously learned skills including drawing nets and finding missing dimensions, surface area, lateral area, and volume of rectangular pyramids and rectangular prisms, including within real-world contexts.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>8.6.1 Warm-Up (~5 min.)</td><td>Would You Rather explanation (SE p. 32)</td><td>8.6.3 Guided Practice (10 min.)</td><td>Students create nets to find surface area and volume of prisms and pyramids (SE p. 34)</td></tr>
<tr><td>8.6.2 Collaboration (10 min.)</td><td>Students collaborate to solve volume and surface area problems (SE p. 32)</td><td>8.6.4 Your Turn! (15 min.)</td><td>Individual practice finding surface area, volume, and missing dimensions (SE p. 35)</td></tr>
<tr><td>8.6.5 Wrap-Up (~5 min.)</td><td colspan="3">Self-reflection on understanding of learning targets and how to overcome a barrier to learning (SE p. 37)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td>Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u> N/A
<u>Additional Terms:</u>
• Lateral face
• Surface area
• Volume</td><td>• Four-function calculator</td><td colspan="2">• Real-world examples to use for comparing volume and surface area may be beneficial.</td></tr>
</table>

Teacher Tips

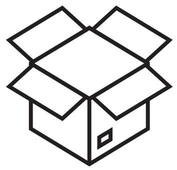
Common Misconceptions	Things to Consider
• Students may attempt to multiply the given volume when prompted to find the missing dimension (rather than dividing or identifying the value of the missing multiple). • Students may get stuck explaining how they would overcome learning barriers from the lesson.	• Students may need a refresher on how to find missing dimensions when given volume for 8.6.2 Collaboration. • Consider having students practice the various vocabulary terms related to three-dimensional figures and finding surface area and volume to foster independence and problem solving. Provide models of prisms and pyramids, then prompt students to recognize the length, width, height, faces, surface area, and volume before 8.6.2 Collaboration. • In some of the real-world contexts the student will not be required to find the total surface area. Students may get confused with the wording “surface area” versus “surface area that should be painted”. Help students understand that in the real-world it is not always necessary to find the area of all of the surfaces.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Compare and Connect Consider using a Compare and Connect routine to stimulate mathematical discussion about surface area and volume. During 8.6.2 Collaboration and 8.6.3 Guided Practice, use the practice problems to have students compare these two mathematical concepts and to connect them to each other through real-world examples and through the lens of prisms versus pyramids.	**MA.K12.MTR.7.1: Real-World Contexts** 8.6.4 Your Turn! engages students in real-world examples on finding surface area and volume with prisms and pyramids. This section highlights math through real-world examples, as students explore how the skills they have learned in this unit translate to real-world scenarios.

Differentiate Instruction

Consider providing a table or similar strategy for graphically organizing information to assist students as they solve the problems for 8.6.2 Collaboration. The table supports students to visually organize given information, partner input, and student calculations throughout the problem-solving process.

	L	W	H	V	SA
Box 1					



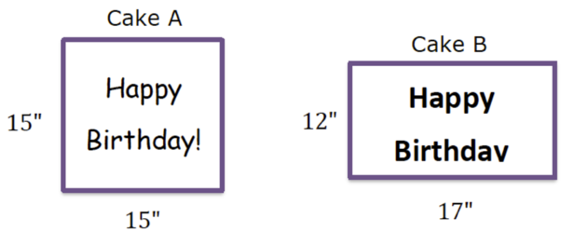
Lesson Notes:

8.6.1 Warm-Up: Would You Rather?

Component Overview: *Students explain why they would rather choose a cake with the indicated dimensions and slices to share. Consider providing students time to connect the real-world scenario to area and compare reasoning through discussion.*

8.6.1 Warm-Up: Would You Rather?

1. Cake A and Cake B are shown below. Cake A is cut into 8 equal slices. Cake B is cut into 6 equal slices. Would you rather have a slice from Cake A or Cake B?



Answers will vary.
I would choose Cake A because it is the biggest.
I would choose Cake B because the slices are bigger.

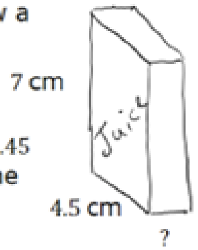
As an alternate method to elicit student responses, post an image of each cake on opposite sides of the room and have the students sit on the side of the room according to which cake they would rather have. Allow them to post their answers and explanations in one location, such as on chart paper, or engage in a class discussion to provide students opportunity to explain their reasoning verbally.

8.6.2 Collaboration: Which Juice Box Is Better?

Component Overview: *Students collaborate to find a missing dimension, then the volume and surface area of two real-world examples of prisms: juice boxes. Have students work with partners to compare the juice boxes, including justifying which prism will use the least amount of material to construct. Consider having students use a table or chart to organize needed information and found measurements. Students can optionally create a poster comparing the two juice boxes, including the organized table of measurements and sketches of the nets/prisms and their work for questions 2 and 4 and reasoning for question 5. Then the posters can be placed around the classroom for a gallery walk.*

8.6.2 Collaboration: Which Juice Box is Better?

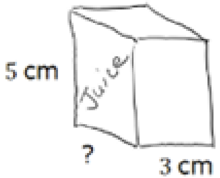
Julius is designing a juice box. He drew a model of his box.



1. If the juice box has a volume of 72.45 cubic centimeters (cm^3), what is the width of the box?
   2.3 cm

2. How much packaging board would Julius need to make a juice box with these dimensions, not taking into consideration any overlap of the faces?
   $115.9 cm^2$

Julius wondered how the juice box would be affected if he changed the dimensions, but kept the volume the same.



3. What is the new length?
   $4.83\ cm$

4. How much packaging board would Julius need to make the new juice box?
   $107.28\ cm^2$

5. If Julius wants to use the least amount of material to construct the juice box, which juice box should he choose? Explain your reasoning.
   He should use the second juice box because the surface area is smaller.

Compare and Connect
Prior to beginning the Collaboration, prompt students to consider how volume and surface area are similar and different. Students may point out that volume involves an amount on the inside of three-dimensional figures and that surface area involves an amount on the outside of three-dimensional figures. Consider providing real-life examples of volume, such as the amount of juice inside of a juice box, and surface area, such as the cardboard that creates the juice box.

Consider providing a table or graphic organizer, such as the table below, for students to use as they problem-solve on question 5. This strategy can be used with any problem in this lesson as an added visual support.

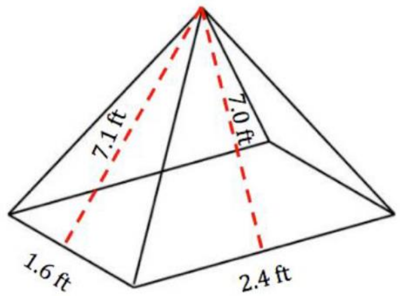
	Length	*Width*	*Height*	*Volume*	*Surface Area*
Juice box 1					
Juice box 2					

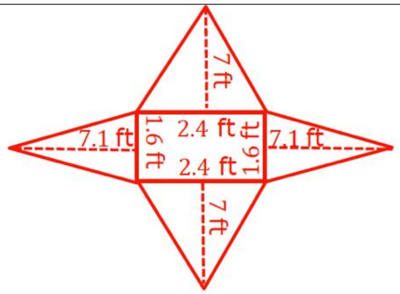
Component Overview: *Students work in partners to practice finding surface area and volume with prisms and pyramids, including identifying the figure to draw a labeled net and needed dimensions. Consider providing chart paper and/or graph paper as students draw the nets and label the measurements.*

8.6.3 Guided Practice: Surface Area and Volume

With your partner, complete the chart for the three-dimensional figures below.

1.



What is the type of figure?	*Pyramid*
Draw a net or the individual faces of the figure. Label the measurements.	
Find the surface area of the figure.	$32\ ft^2$

As partnerships begin working on these problems, pose these questions to foster strategic math reasoning and various approaches to finding surface area and volume of different figures:

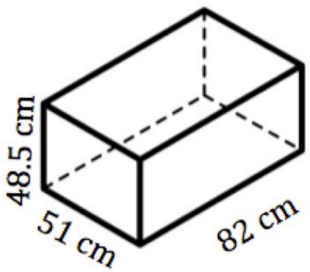
- *What are the differences between rectangular pyramids and rectangular prisms?*
- *How do the differences between pyramids and prisms affect how you find the volume? The surface area?*
- *What different strategies do you know to find the surface area/volume of a prism/pyramid? Which strategy makes the most sense for this problem? Why?*

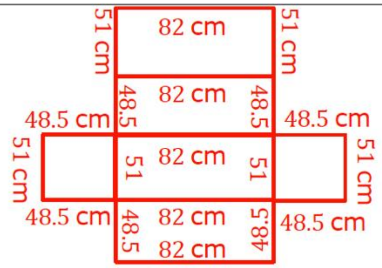
Consider providing students with chart paper to complete the tables and draw the nets for questions 1 and 2.

Having ample space to draw a net is important as students solidify the visual representation of a three-dimensional figure as a collection of two-dimensional shapes. Seeing both figures from questions 1 and 2 on the chart paper can help strengthen conceptual understanding as students compare and connect the figures and approaches to finding surface area and volume.

8.6.3 Guided Practice: Surface Area and Volume (continued)

2.



What is the type of figure?	*Rectangular Prism*
Draw a net of the figure or the individual faces. Label the measurements.	
Find the surface area of the figure.	$21{,}265\ cm^2$
Find the volume of the figure.	$V = 51\ \times 82\ \times 48.5\ = 202{,}827\ cm^3$

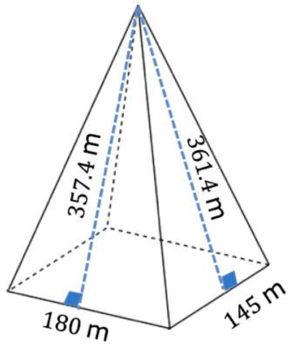
Prompt students to find an example of a rectangular prism or pyramid in the classroom and complete a table like question 2 as additional practice.

8.6.4 Your Turn!

Component Overview: *Students individually find volume and surface area by applying their understanding to four real-world problems related to rectangular prisms and pyramids. As students work on these problems, consider running a small group to provide additional support to students who may still have difficulty finding missing dimensions, creating nets, and/or finding surface area and volume.*

8.6.4 Your Turn!

1. Nikolas is constructing a skyscraper with walls made of only glass that is 350 meters (m) high and is in the shape of a rectangular pyramid.



   a. What is the surface area of the skyscraper?
   142,835 m^2

   b. Using the model shown, determine how much of the surface area will be made of glass.
   116,735 m^2

For question 1, it may be helpful to first have visual students draw a net to represent the pyramid. This practice may help students recognize how the lateral triangular faces have two different heights and how many faces there will be with those measurements.

For question 1b, students without background knowledge on skyscrapers may not recognize which sections of the skyscraper will be covered in glass. Connect mathematics to the real-world by showing a picture of a pyramid-shaped skyscraper and asking:

- *What do you notice about the picture? Which parts have glass?*
- *Which parts of our model will not have glass? How do you know?*
- *How can you use this knowledge to find the amount of surface area that will be covered in glass?*

8.6.4 Your Turn! (continued)

2. A cooler has a length of 78 centimeters (cm), a height of 74 cm, and a width of 40 cm.

74 cm

40 cm

78 cm

   a. Find the surface area of the cooler.
   $23{,}704\ cm^2$

   b. Find the volume of the cooler.
   $V = 78\ \times 40\ \times 74\ = 230{,}880\ cm^3$

   c. If the **inside** of the cooler has a length 62 cm, a height of 58 cm, and a width of 28 cm, how much smaller is the volume of the inside of the cooler than the volume of the entire cooler?

58 cm

28 cm

62 cm

$$V_{inside} = 62\ \times 28\ \times 58\ = 100{,}688\ cm^3$$
$$V_{entire} - V_{inside} = 230{,}880 cm^3 - 100{,}688 cm^3$$
$$= 130{,}192\ cm^3$$

Consider bringing a cooler to class for students to analyze and manipulate. Though the dimensions between the model and actual cooler will be different, students can visualize how the inside and outside of the cooler have different dimensions because of the thickness of the sides, which may aid problem-solving for question 2.

Compare and Connect
For question 2c, students may notice that the question is not just asking for the volume of the inside of the cooler, but it is also asking for the difference between the two volumes. As students work individually or during small group discussions, students can recognize different approaches to comparing the volumes, including finding the volume of the inside of the cooler and subtracting it from the volume of the outside of the cooler.

8.6.4 Your Turn! (continued)

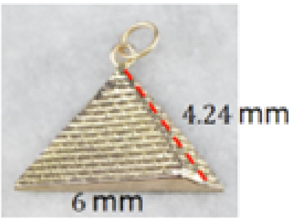
3. Tamya is creating a gardening box that needs to have a volume of 13.5 ft^3.

   a. If the box has a width of 3 feet and a length of 2 feet, what is the height?

   $h = 2.25\ ft$

   b. Tamya decided to paint the exterior walls of the gardening box and needs to know how much paint to buy. What is the surface area of the area she wants to paint?
   $22.5\ ft^2$

4. Ryan makes custom jewelry and recently received an order for a pendant in the shape of a rectangular pyramid. Ryan designed a hollow gold pendant to have a square base with a side length of 6 millimeters (mm) and a slant height of 4.24 mm.



   How much gold does Ryan need to make the pendant?
   $86.88\ mm^2$

As an optional hands-on extension, prompt students to recreate a life-size model of one of the prisms or pyramids from questions 2, 3, or 4. This kinesthetic practice will not only further understanding in creating accurate nets, but also in measurement, as students need to use a ruler to measure precisely. Students can use cardstock paper, tape, modeling clay, or other classroom models to build their representations.

8.6.5 Wrap-Up: Reflection

Component Overview: *Students rate their conceptual understanding of the lesson's learning targets using a number scale, then describe a barrier to learning from today's lesson and a way they will try to overcome it.*

8.6.5 Wrap-Up: Reflection

1. Rate yourself on each of the concepts.

   4 – I can do this on my own.
   3 – I can do this but need more practice.
   2 – I need help getting started.
   1 – I do not understand yet.

Concept	Rating
I can find the surface area of a right rectangular pyramid using a net or the figure.	1 2 3 4
I can find the surface area of a right rectangular prism using a net or the figure.	1 2 3 4
I can find the volume of a right rectangular prism.	1 2 3 4

2. A barrier to my learning today was …
   Answers will vary.
   Trying to remember how to find the area of a triangle.
   Trying to remember how many triangles a pyramid has.

3. I will try to overcome this by …
   Answers will vary.
   Remembering that the area of a triangle is half of a rectangle.
   Remembering the number of triangles on a pyramid is determined by the rectangular base which has 4 sides.

For students who get stuck on question 3, provide ideas for ways students can overcome barriers to learning by writing a list on the whiteboard of shared ideas from the class.

Examples:
- *Texting with my friend about question 2 when I get home*
- *Watching the Study Expert videos after dinner to try to learn from them*
- *Attending tutoring on __________ (day)*